CORE MATHEMATICS GRADE 10: AREA OF A PART OF A CIRCLE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.6 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.6: Area of a Part of a Circle
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Area of a circle (review) – Area of an annulus (ring between two concentric circles) – Area of a sector of a circle – Area of an annular sector – Area of a segment of a circle – Area of a common region between two intersecting circles – Application of circle-part areas to real-life situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the area of an annulus in different situations, b) determine the area of a sector of a circle in different situations, c) determine the area of an annular sector in different situations, d) determine the area of a segment of a circle in different situations, e) determine the area of a common region between two intersecting circles, f) apply areas of parts of a circle to real-life situations, g) appreciate the use of areas of parts of a circle in real-life situations.
Core Competencies	– Critical Thinking and Problem Solving: the learner analyses real-life objects (e.g., a Kenyan chapati pan, a circular shamba plot, the Nairobi roundabout) and decomposes their regions into annuli, sectors, segments and intersecting-circle areas to compute usable space or material needed. – Communication and Collaboration: the learner discusses in groups how to partition circular cut-outs into parts, justifies formulae derivation steps, and presents findings to peers. – Creativity and Imagination: the learner designs and improvises models of circular regions (e.g., paper cut-outs of mandazi cutters, circular irrigation plots) to demonstrate and verify formulae for parts of a circle. – Digital Literacy: the learner uses calculators and, where available, digital geometry tools to compute and verify areas of parts of circles efficiently.
Core Values	– Responsibility: the learner carefully handles rulers, compasses, calculators and circular cut-out materials, and accurately records measurements to ensure correct area computations.

	– Integrity: the learner honestly records calculated values and verifies answers against physical measurements, accepting and correcting errors rather than falsifying results. – Unity: the learner works collaboratively in groups to derive and apply formulae, respecting every member's contribution during discussions and model-building activities.
Science & Engineering Practices	– Using Mathematics and Computational Thinking: learners apply formulae for annulus, sector, annular sector, segment, and intersecting-circle regions — substituting measured values, manipulating equations, and interpreting numerical results in context. – Developing and Using Models: learners build physical and diagrammatic models (circular cut-outs, shaded region sketches, scale diagrams of irrigation plots) to represent and reason about areas of parts of circles. – Asking Questions and Defining Problems: learners pose and refine questions about why different parts of the same circle have different areas, driving investigation of each sub-region formula. – Constructing Explanations: learners articulate, in writing and orally, how each formula is derived from the whole-circle area, connecting the fraction of the full angle (360°) to the fraction of the total area.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development (Environmental Education): the learner applies area-of-circle-parts to calculate usable land in circular irrigation schemes common in the Rift Valley and arid regions of Kenya, reinforcing efficient land and water use. – Social Cohesion: the learner works in mixed-ability groups to measure, compute, and verify areas, building respect and teamwork across diverse backgrounds. – Financial Literacy: the learner estimates material costs for circular structures (e.g., tiling a circular stage at a Nairobi school, fencing an annular flower bed) by computing exact areas, linking mathematics to budgeting and enterprise. – Road Safety: the learner computes the area of circular roundabouts and annular road markings, connecting geometry to road-design awareness.
Career Connections	– Architecture and Civil Engineering: professionals design circular stadiums, roundabouts, and curved building features requiring precise area calculations of sectors and annuli. – Land Survey and Cartography: surveyors in Kenya's National Land Commission compute circular and part-circular plot areas for titling and valuation. – Agriculture and Irrigation Engineering: engineers designing centre-pivot irrigation schemes (common in Laikipia and Nakuru counties) use sector and annulus areas to plan water coverage and crop layout. – Graphic Design and Visual Arts: designers creating circular logos, Kanga fabric patterns, and decorative tilework use sector and segment geometry. – Medicine and Radiology: medical imaging professionals interpret circular cross-sections of organs, requiring understanding of circular region areas. – Manufacturing and Artisanal Craft: jua kali artisans cutting circular metal sheets (e.g., sufuria lids, chapati pans) minimise waste by calculating segment and annular cut areas.
Focus for Lessons	This unit extends learners' understanding of circle geometry by investigating the areas of distinct parts of a circle — the annulus, sector, annular sector, segment, and the common region between two intersecting circles. Using Kenya-relevant contexts such as circular irrigation plots in Laikipia, Nairobi roundabout road markings, and the cross-sections of sufuria and chapati pans, learners derive each formula from first principles before applying it to real-life problems. The unit balances hands-on measurement with algebraic reasoning, building both procedural fluency and conceptual understanding aligned to CBC competencies.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?

	KICD KEY INQUIRY QUESTIONS: 1. How do we determine the area of parts of a circle? 2. How are areas of parts of a circle applied in real-life situations?
Anchoring Phenomenon	A farmer in Laikipia County uses a centre-pivot irrigation system: a long pipe rotates around a central water source, watering a perfectly circular area of maize. The farmer wants to expand the system by adding a second outer pipe ring, and also needs to know exactly how much of the field is watered when the pipe has only rotated part-way around. When learners examine an aerial photograph of the farm, they notice that the watered regions form rings, wedge-shaped slices, and overlapping patches — yet the farmer's title deed gives only the area of the full circular plot. Students are puzzled: the full-circle formula (πr^2) gives the total area, but how do we find the area of just the ring between the two pipes, or the wedge-shaped slice the pivot has swept so far, or the region where two adjacent circular plots overlap? The anchoring phenomenon drives learners to decompose circles into precise sub-regions and derive formulae for each, ultimately helping the farmer plan planting, fencing, and water budgets accurately.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Area of an Annulus: Learners examine the Laikipia irrigation photograph and identify the ring-shaped watered zone. They review the full-circle area formula, then derive the annulus formula $A = \pi(R^2 - r^2)$ using concentric circular cut-outs. They solve problems involving annular irrigation rings and chapati-pan offcuts. Lesson 2 — Area of a Sector: Learners connect the irrigation pivot's partial sweep to a sector ("pie slice"). They derive $A = (\theta/360) \times \pi r^2$ by comparing sectors of different angles cut from the same paper circle. They apply the formula to calculate how much maize area has been watered after a given angle of rotation, and to the Kisumu roundabout crossing zone. Lesson 3 — Arc Length and Sector Perimeter (Supporting Sector Work): Learners establish arc length $l = (\theta/360) \times 2\pi r$ to support later segment work, then calculate the full perimeter of sectors. Contextualised around fencing a sector-shaped school garden in Nakuru. Lesson 4 — Area of an Annular Sector: Learners combine lessons 1 and 2 to derive the annular sector formula $A = (\theta/360) \times \pi(R^2 - r^2)$. They apply it to the outer irrigation ring swept through a partial angle, and to the painted road-marking problem in Kisumu. Lesson 5 — Area of a Segment of a Circle: Learners derive the segment formula $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ by subtracting the triangle area from the sector area. They apply it to the Turkana borehole cover hatch problem and to calculating the area of a chord-cut region on a circular sufuria lid. Lesson 6 — Area of a Common Region Between Two Intersecting Circles: Learners use two overlapping circular cut-outs (the Kericho tea-tray stencils) to identify the lens-shaped common region. They derive its area as the sum of two circular segments and apply it to overlapping irrigation plots and the school art project stencil. Lesson 7 — Mixed Application and Problem Solving: Learners tackle multi-step real-life problems combining two or more circle-part formulae (e.g., an annular sector with a segment cut out; a composite shape involving a full circle and an intersecting region). Problems are set in Nairobi, Mombasa, and Rift Valley contexts. Learners work in groups, present solutions, and peer-assess accuracy. Lesson 8 — Assessment, Model Revision and Consolidation: Learners complete a class written test and then revise their cumulative diagrammatic model of the circle decomposed into all learned sub-regions, annotating each with its formula and a real-

	life Kenyan application. Groups share final models; teacher consolidates key formulae and links back to the driving question and anchoring phenomenon.
Supporting Phenomena	– A jua kali artisan in Kamukunji, Nairobi, cuts circular chapati-pan lids from a flat metal sheet and needs to know the area of the leftover ring-shaped offcut (annulus) to minimise material waste. – A road engineer marks a painted annular ring around a roundabout in Kisumu City and must calculate the exact area of paint required for the ring and its sector-shaped pedestrian crossing zones. – A student in Kericho notices that two overlapping circular tea-tray stencils create a leaf-shaped common region and wonders how to calculate its exact area for a school art project. – A borehole technician in Turkana digs a circular well and installs a concrete annular cover; they need the area of the concrete segment left after cutting an access hatch, to order the right amount of cement.

LESSON 1 (80 minutes): Anchoring Phenomenon & Area of an Annulus	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and derive the formula for the area of an annulus, connecting it directly to the Laikipia centre-pivot irrigation context.
Knowledge	Learners recall the full-circle area formula $A = \pi r^2$, identify the annulus as the region between two concentric circles, and derive the annulus formula $A = \pi(R^2 - r^2)$.
Skills	Learners measure radii of concentric circular cut-outs, subtract areas algebraically, record observations systematically, and apply the annulus formula to solve real-life problems involving irrigation rings and chapati-pan offcuts.
Attitudes	Learners appreciate the value of precise mathematical calculation in avoiding waste and supporting Kenyan farmers and artisans, and collaborate respectfully in group investigations.
Key Inquiry Question	How do we determine the area of parts of a circle?
Purpose in Storyline	This is the opening lesson of the sub-strand. Learners encounter the Laikipia centre-pivot irrigation photograph for the first time, notice that the full-circle formula cannot answer the farmer's questions about ring-shaped watered zones, and take their first step toward decomposing a circle into precise sub-regions by deriving the annulus formula $A = \pi(R^2 - r^2)$. Every subsequent lesson will build on this anchoring moment.
Safety Notes	Scissors and compasses are used during the cut-out activity. Learners must handle pointed instruments with care, keep compasses closed when not in use, and pass scissors handle-first. Supervise closely during cutting. Ensure desks are clear of bags to avoid trip hazards during group movement.

B. LESSON OVERVIEW

Learners open with a striking aerial photograph of a Laikipia centre-pivot irrigation farm and immediately notice ring-shaped, wedge-shaped, and overlapping watered zones that the full-circle formula cannot explain. After recording initial predictions about how to find the ring area, they conduct a hands-on concentric circular cut-out investigation to derive $A = \pi(R^2 - r^2)$ for an annulus. They apply the formula to two Kenyan real-life contexts — an irrigation ring and a chapati-pan offcut — and open the Driving Question Board (DQB) by posting questions about the other shapes they noticed in the photograph. The lesson establishes the phenomenon as the throughline for all subsequent lessons in the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Magnetic field created by a current carrying]()	Learners receive a printed aerial photograph (or projected image) of the Laikipia centre-pivot irrigation farm showing two concentric watered rings, a partially-swept	Display the photograph prominently. Say: 'Look carefully at this farm in Laikipia County. A farmer here uses a centre-pivot irrigation system — a long pipe	Predict-before-instruction (anticipatory brainstorm). By committing predictions to paper before any instruction, learners activate prior knowledge of the	Observable indicator: every learner has written at least one recognisable shape name and attempted a written reason for whether $A = \pi r^2$ is sufficient.

wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	wedge, and an overlapping patch where a neighbouring plot's system encroaches. Working individually for 3 minutes, each learner writes answers to three prompt questions on a sticky note: (1) What shapes do you see in the watered area? (2) The farmer's title deed says the full circular plot has radius 120 m — can you use $A = \pi r^2$ to find the area of just the ring between the two pipes? Why or why not? (3) Estimate: if the outer pipe is 120 m from the centre and the inner pipe is 80 m, roughly what fraction of the total circle is the ring? Learners then share predictions with a shoulder partner for 2 minutes before the teacher takes whole-class cold calls.	that rotates around a central water source. What do you notice? What do you wonder?' Allow 10 seconds of silent observation (WAIT TIME). Then: 'Write your own answers to the three questions on your sticky note. You have 3 minutes — do not confer yet.' Circulate silently, noting range of predictions. After partner share, cold-call exactly 4 learners: one who attempted a numerical estimate, one who said $A = \pi r^2$ is sufficient, one who said it is not, and one who named a shape other than a ring. Record key prediction phrases verbatim on the board under the heading 'Our Predictions'. Do NOT correct errors yet. Say: 'Hold these ideas — by the end of today you will know which of these are right, and you will have new questions too.'	circle area formula, surface misconceptions (e.g. 'just subtract the radii'), and generate genuine curiosity about why the full-circle formula is insufficient. Recording predictions publicly creates accountability and a reference point for revision.	Teacher scans sticky notes during circulation; if more than one-third of learners leave question 2 blank, pause and prompt: 'Think about what information $A = \pi r^2$ gives you — does it give you the area of just part of the circle?' Collect sticky notes at end of phase to compare with end-of-lesson exit tickets.
Observe Phase 📹 **VIDEO:** Magnetic field created by a current carrying wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In groups of four, learners conduct a concentric circular cut-out investigation. Each group receives: two pre-drawn concentric circles on card (outer radius R = 12 cm, inner radius r = 8 cm), scissors, a ruler, and a calculator. Step 1 — Measure: learners measure and record R and r. Step 2 — Cut: learners cut out the full outer circle, then cut out the inner circle, producing a ring-shaped piece (the annulus) and a small disc. Step 3 — Record in a table: they calculate (a) area of full outer circle πR^2, (b) area of inner disc πr^2, and (c) area of the ring by physically setting aside the disc and writing $Area\_ring = Area\_outer - Area\_inner$. Step 4 — Generalise: groups repeat the calculation using a second set of values (R = 10 cm, r = 6 cm) drawn on paper, without cutting, to confirm the pattern. Step 5 — Connect to	Distribute materials and say: 'You are going to discover the formula the farmer needs by doing what mathematicians do — investigate with physical objects, then generalise.' Demonstrate measuring the radius of the outer circle only (do not demonstrate the subtraction). Circulate every 90 seconds. At the cutting step, ask: 'What does the ring-shaped piece represent on the Laikipia farm?' (WAIT TIME 12 seconds) Cold-call 2 groups. At the recording step, ask: 'How did you get the area of the ring — what operation did you use?' (WAIT TIME 10 seconds) Cold-call 3 groups. If a group writes $Area\_ring = \pi(R - r)^2$, redirect with: 'Test your formula with R = 12, r = 8 on your calculator. Does your answer match what you get by subtracting the two separate areas? Investigate why.' After Step 4, say:	Concrete-Representational-Abstract (CRA) progression. Learners move from physical card cut-outs (concrete) to labelled diagrams connecting cut-outs to the photograph (representational) to a generalised algebraic pattern (abstract). The physical act of removing the inner disc makes the subtraction structure viscerally obvious, grounding the algebra in a tangible experience before symbols are introduced.	Observable indicator: each group's recording table shows correct numerical values for πR^2 and πr^2 (using $\pi \approx 3.14$ or the π key) and writes the ring area as a difference, not a sum or product. Teacher checks at least three tables during circulation. Red flag: any group writing $Area\_ring = \pi(R - r)^2$ — redirect immediately using the test-with-numbers prompt. Green flag: group spontaneously factors the expression as $\pi(R^2 - r^2) = \pi(R+r)(R-r)$ — note for Phase 3 extension.

	phenomenon: groups return to the Laikipia photograph and label which part of the irrigation system corresponds to R and which to r.	'Write one sentence that describes the pattern you see every time.' Cold-call 2 groups to share sentences; scribe the most precise one on the board.		
Explain Phase 📹 **VIDEO:** Magnetic field created by a current carrying wire Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole class reconvenes. The teacher formalises the annulus formula from the groups' sentences, then learners apply it in two contextualised problems worked first individually (5 min) then compared with a partner. Problem 1 (Irrigation ring): The Laikipia farmer's outer pipe sweeps a radius of 120 m and the inner pipe radius is 80 m. Calculate the area of the annular ring that is watered. Give your answer in hectares (1 ha = 10 000 m^2). Problem 2 (Chapati pan): A chapati pan has an outer radius of 18 cm. A smaller inner ring of radius 11 cm is the heated cooking surface, and the outer border stays cool. What is the area of the cool border ring? How many such border rings could be cut from a circular sheet of dough of radius 18 cm if the inner dough circle of radius 11 cm is used for mandazi? Learners write full solutions showing substitution into $A = \pi(R^2 - r^2)$, then connect each answer back to the phenomenon: 'How does this help the Laikipia farmer?'	Begin by asking: 'Who can give me one sentence that captures what we found?' (WAIT TIME 12 seconds) Cold-call the group whose sentence was scribed. Write on board: 'Area of annulus = Area of outer circle − Area of inner circle = $\pi R^2 - \pi r^2 = \pi(R^2 - r^2)$.' Say: 'This is the formula. Notice we can factorise: $\pi(R - r)(R + r)$. We will explore why that factorisation is useful later in the unit.' Do NOT spend more than 2 minutes on the factorisation — flag it for a future lesson. Display Problem 1. Say: 'Work this out individually. You have 5 minutes. Show every substitution step.' After 5 minutes, say: 'Compare your answer with your partner. If you disagree, work out where the difference began.' Cold-call 2 pairs to present solutions; invite the class to agree or challenge. Repeat for Problem 2. Close by asking: 'How many hectares of maize can the farmer water with just the ring? How does knowing this help with planning fertiliser or seed quantities?' (WAIT TIME 15 seconds) Take 3 responses. Say: 'You have just solved a real problem that a Laikipia farmer faces. Tomorrow we will look at the wedge-shaped region the pivot sweeps — that needs a different approach.'	Contextualised problem solving with phenomenon anchoring. By situating both problems explicitly in the Laikipia farm and a familiar Kenyan kitchen context (chapati pan), learners see mathematics as a tool for real decisions, not an abstract exercise. Requiring learners to write a connecting sentence ('How does this help the farmer?') forces them to close the loop between formula and phenomenon, deepening transfer.	Observable indicator: learner solutions to Problem 1 show correct substitution ($\pi \times (120^2 - 80^2) = \pi \times (14400 - 6400) = \pi \times 8000 \approx 25\ 133\ m^2 \approx 2.51$ ha) with units clearly stated. Problem 2 solutions show $A = \pi(18^2 - 11^2) = \pi(324 - 121) = 203\pi \approx 637.7\ cm^2$. Teacher collects four books at random during pair-comparison to check substitution and unit conversion. Red flag: learners subtracting radii before squaring — address immediately with a whole-class 'spot the error' exercise using a deliberate wrong answer on the board.
Driving Question Board (DQB) Creation	Learners return to their original sticky-note predictions from Phase 1. Each learner reads their prediction and decides: (a) 'I now	Say: 'Look at what you wrote at the start of the lesson. Has anything changed in your thinking?' (WAIT TIME 10 seconds) 'Move your	Driving Question Board (DQB) as a public knowledge-tracking tool. The DQB externalises metacognition — learners see	Observable indicator: every learner moves at least one sticky note and adds or stars at least one question. Teacher checks that

📹 **VIDEO:** [Magnetic field created by a current carrying wire] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	understand this' — sticky note moves to the 'Explained' column; (b) 'This is still a question' — sticky note stays on the board; or (c) 'This gave me a NEW question' — learner writes a new sticky note. The class then does a gallery walk (3 minutes) around the DQB, adding star stickers to questions they share. The teacher reads out the top-starred questions, groups them by theme (e.g. wedge/sector shapes, overlapping regions, rings), and writes the overarching driving question at the top of the board: 'How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?' Teacher labels three columns on the DQB: 'Annulus ✓ (today)', 'Sector/Segment — coming soon', 'Overlapping circles — coming soon'.	sticky notes as instructed and add any new questions — 2 minutes.' After the gallery walk, read the top three starred questions aloud: e.g. 'How do we find the area of the wedge the pipe sweeps?' / 'What about where two farms overlap?' / 'Does the formula change if the circles are not centred at the same point?' Say: 'These are excellent mathematical questions. Over the next lessons, each one will get answered. We are building a map of our own understanding.' Pin the driving question card prominently. Say: 'Every lesson, we will return here and tick off what we have learned.' Photograph the DQB for the ARES platform record.	visually which aspects of the phenomenon remain unexplained, creating productive tension that motivates subsequent lessons. The act of categorising questions by theme mirrors scientific practice of organising inquiry and signals to learners that their questions, not just the teacher's agenda, drive the unit.	questions in the 'still open' column include references to sectors, segments, or overlapping circles — these are essential for the storyline to proceed. If the DQB contains only annulus questions, prompt: 'Look back at the photograph — what other shapes did you see besides rings? What questions do those shapes raise?'
Model Building Phase 📹 **VIDEO:** [Magnetic field created by a current carrying wire] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their mathematics notebook to a fresh double page headed 'My Circle Parts Model — Lesson 1'. They draw a large circle and label: (i) a concentric inner circle with radius r and outer radius R; (ii) the annulus region shaded in green; (iii) the formula $A\_annulus = \pi(R^2 - r^2)$ with an arrow pointing to the shaded region. Below the diagram they write: 'What I can explain: the area of the ring-shaped (annular) region between two concentric circles. What I still need: a way to find the area of the wedge the pivot sweeps, and the area where two plots overlap.' Finally, learners complete a 3-2-1 exit ticket: 3 things observed in the photograph, 2 things learned today, 1 question still on their mind. Exit tickets are	Say: 'We are going to start building a model — a growing diagram in your notebook — that will get more complete every lesson as we answer more of our driving question. Today you can only fill in one part: the annulus.' Demonstrate on the board: draw the two concentric circles, shade the annulus, write the formula. Say: 'This model will grow. By the end of the sub-strand, you will have a full diagram showing every part of a circle with its formula.' Circulate for 4 minutes, checking that: (a) both circles are labelled with R and r, (b) the formula is written correctly with brackets, (c) the 'what I still need' section names at least one other shape. Collect exit tickets at the door. Say: 'Your questions on those	Cumulative annotated model (notebook diagram). Building a single growing diagram across all lessons in the sub-strand provides a persistent visual anchor that makes the decomposition of a circle into sub-regions explicit and cumulative. Learners can literally see what they know and what is missing, which supports self-regulated learning. The 3-2-1 exit ticket provides immediate formative data aligned to the three lesson goals: observation, knowledge, and ongoing inquiry.	Observable indicator: notebook diagrams show both circles labelled R and r, annulus shaded, and formula $A = \pi(R^2 - r^2)$ correctly written. Exit ticket '1 question' items should reference shapes beyond the annulus (sector, segment, or overlapping region). Red flag: learners writing the formula as $\pi(R - r)^2$ in the model — intercept before they leave and ask them to test with numbers from Problem 1 to self-correct. Green flag: learners who spontaneously write $A = \pi(R+r)(R-r)$ — invite them to explain the factorisation to the class at the start of Lesson 2.

	handed to the teacher as learners leave.	tickets will shape what we do next lesson.'		

D. TEACHER REFLECTION

After the lesson, review the Phase 1 sticky notes against the Phase 5 exit tickets and ask: (1) Did learners' initial predictions reveal a specific misconception (e.g. subtracting radii before squaring) that resurfaced in the Explain phase? If so, plan an explicit 'spot-the-error' warm-up for Lesson 2. (2) Were the DQB questions rich enough to span sectors, segments, and overlapping circles — the three remaining storyline threads? If the DQB is thin, bring a second photograph (e.g. a Nairobi roundabout aerial view showing overlapping circular patterns) to Lesson 2 to broaden the phenomenon. (3) Did all groups successfully complete the cut-out investigation within the allotted time? If not, consider pre-cutting the inner circles for lower-dexterity groups in future iterations. (4) Check whether unit conversion (m^2 to hectares) caused errors in Problem 1 — if more than one-third of learners made this error, open Lesson 2 with a 5-minute unit-conversion drill using Kenyan land measurement contexts (e.g. shamba sizes in Kericho or Rift Valley). (5) Note which learners were not cold-called today and prioritise them in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	An aerial photograph of a Laikipia County centre-pivot irrigation farm showing ring-shaped (annular), wedge-shaped, and overlapping watered zones that the full-circle formula $A = \pi r^2$ cannot fully explain.
What did I learn?	The area of an annulus (the ring between two concentric circles of outer radius R and inner radius r) is found by subtracting the inner circle area from the outer circle area: $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$. This formula was derived using concentric circular card cut-outs and applied to an irrigation ring in Laikipia and a chapati-pan border.
How does this explain the phenomenon?	Why $A = \pi r^2$ alone is insufficient for the farmer's planning needs, and how the annulus formula $A = \pi(R^2 - r^2)$ gives the exact area of the ring-shaped watered zone, enabling accurate calculation of planting area, fencing, and water budgets for the ring section of the farm.

LESSON 2 (80 minutes): Area of a Sector: The Pivot's Sweep	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive and apply the formula for the area of a sector of a circle in real-life contexts.
Knowledge	Learners will know that a sector is a 'pie-slice' region of a circle bounded by two radii and an arc, and that its area is a fraction of the full circle's area given by $A = (\theta/360) \times \pi r^2$.
Skills	Learners will be able to: (i) identify and name sectors in real-world diagrams; (ii) derive the sector area formula by comparing sectors of different angles cut from the same paper circle; (iii) calculate the area of a sector given the radius and angle; (iv) apply the formula to the Laikipia centre-pivot irrigation context and the Kisumu roundabout crossing zone.
Attitudes	Learners will appreciate that precise mathematical formulae enable Kenyan farmers, engineers, and urban planners to make accurate decisions that reduce waste and support sustainable development.
Key Inquiry Question	How do we determine the area of parts of a circle, and how are areas of parts of a circle applied in real-life situations?
Purpose in Storyline	In Lesson 1, learners established the area of a full circle (πr^2) and identified the annulus (ring between two pivot pipes). Today they zoom in on the moment the pivot pipe has only rotated part-way around the Laikipia field — the watered region is a wedge, a sector. By deriving $A = (\theta/360) \times \pi r^2$, learners gain the second tool they need to answer the farmer's question: exactly how much maize has been watered after a given angle of rotation?
Safety Notes	Scissors are used during the paper-cutting activity. Learners must cut away from their body and keep scissors flat on the desk when not in use. Ensure adequate desk space between pairs.

B. LESSON OVERVIEW
Learners begin by predicting what fraction of a circular maize field has been watered after the centre-pivot pipe rotates through a given angle. Through a structured paper-cutting investigation, they cut sectors of angles 90°, 120°, 60°, and 45° from identical paper circles and compare their areas as fractions of the whole, inductively building the formula $A = (\theta/360) \times \pi r^2$. They then apply the formula to two Kenyan contexts: the Laikipia irrigation pivot and a pedestrian crossing arc at the Kisumu Mega roundabout. The lesson closes with learners updating the class Driving Question Board and revising their circle-decomposition model to include the sector.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English	Each learner receives a printed aerial-view sketch of the Laikipia centre-pivot field (a full circle of radius 40 m). The sketch shows the pivot pipe frozen at 120° of rotation from its start position, with the swept region shaded. Learners	Distribute the Laikipia sketch handout. Say exactly: 'The farmer's pivot pipe started at the boundary and has swept 120 degrees — one third of the way around. Before I teach you anything, I want your best	Prediction with Anchor Phenomenon — activating prior knowledge of fractions and circle area (from Lesson 1) and creating cognitive dissonance that motivates the investigation. Writing predictions before instruction	Circulate and scan exercise books during individual writing. Observable indicator: learner has written a fraction (not just a verbal guess) AND connected it to 360° or to πr^2 in some form. Flag learners who leave the expression

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	individually write answers to two prompt questions in their exercise books: (1) 'What fraction of the full circular field do you think has been watered? Write it as a fraction and explain your reasoning.' (2) 'If the full field covers an area of πr^2, write an expression you think might give the area of just the shaded wedge.' Learners are NOT shown the formula yet. After 4 minutes of individual thinking, they share their prediction with a shoulder partner for 2 minutes.	mathematical guess. Use what you already know about fractions and circles.' Set a visible 4-minute timer. Circulate silently — do NOT confirm or correct. After 4 minutes say: 'Share your prediction with the person next to you. You have 2 minutes.' After partner share, cold-call 3 learners (select one who wrote a fraction, one who wrote an algebraic expression, one who drew a diagram). For each, ask: 'What evidence from the diagram made you think that?' Allow 10 seconds WAIT TIME after each question before taking the response. Record the three predictions verbatim on the board under the heading 'OUR PREDICTIONS — we will test these today.'	ensures learners have a stake in the outcome.	blank — pair them with a stronger predictor during the investigation phase.
Observe Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In groups of four, learners receive: four identical paper circles (diameter 20 cm, pre-drawn), a protractor, scissors, a ruler, and a recording table printed on a worksheet. Each group member cuts one sector from their circle at a different angle: 90°, 120°, 60°, and 45°. They then: (a) express each sector angle as a fraction of 360°; (b) calculate the area of the full circle ($\pi \times 10^2 = 314.16$ cm²); (c) multiply the fraction by the full-circle area to predict the sector area; (d) physically stack or fold sectors to verify relative sizes are consistent with their fractions; (e) complete the recording table with columns: Angle θ | θ/360 | Full circle area | Predicted sector area = $(\theta/360) \times \pi r^2$. The teacher then projects a digital simulation (or draws on the board) showing how as θ increases from 0° to 360° the sector area grows proportionally, reinforcing linearity.	Before groups start cutting, say: 'This investigation mirrors exactly what happens on the Laikipia farm — the pivot pipe sweeps out a sector. Your job is to find the mathematical pattern.' Demonstrate how to use a protractor to mark the sector angle accurately from the circle's centre — do this once slowly for the whole class. Set a 20-minute timer. At the 10-minute mark, circulate and ask each group: 'What pattern are you noticing between the angle and the area?' — allow 10 seconds WAIT TIME. At the 18-minute mark, bring the class back together. Cold-call 2 groups to share their completed table. Ask the whole class: 'If I told you the angle was 1 degree, what would the sector area be?' (WAIT TIME 12 seconds.) Then ask: 'What if the angle was 360 degrees?' (WAIT TIME 10 seconds.) Guide learners to	Inductive Pattern Recognition through Manipulatives — learners generate data from physical paper sectors before seeing the formula. This mirrors the NGSS Science and Engineering Practice of 'Analysing and Interpreting Data' and 'Using Mathematics and Computational Thinking.' The physical stacking of sectors makes the proportional relationship tangible.	Collect one recording table per group at the end of the phase. Observable indicator: all four rows of the table are completed with correct θ/360 fractions AND correct sector area values (90°→78.54 cm²; 120°→104.72 cm²; 60°→52.36 cm²; 45°→39.27 cm²). Groups with two or more errors receive targeted re-teaching in Phase 3.

		articulate: 'The area of the sector = (angle/360) × area of the full circle.' Write the generalised formula on the board: $A = (\theta/360) \times \pi r^2$. Point back to the prediction board and ask: 'Whose prediction was closest? Why?'		
Explain Phase 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formalises the derivation on the board, connecting each step to the paper-circle investigation. Learners copy the derivation into their notes. They then work through three graded application problems individually: (1) LAIKIPIA PIVOT — The centre-pivot pipe has a radius of 40 m and has rotated through 120°. Calculate the area of maize that has been watered. (2) KISUMU ROUNDABOUT — A pedestrian refuge island at the Kisumu Mega roundabout is shaped like a sector with radius 6 m and angle 75°. Calculate its area to help the county engineer estimate the amount of paving material needed. (3) EXTENSION — A Kericho tea farmer uses a sector-shaped plot with area 1 256 m^2. If the radius is 40 m, find the angle of the sector. Learners who finish early attempt the extension problem. The class reviews problems 1 and 2 together, with learners presenting solutions on the board.	Begin the formal derivation by pointing to the paper-circle table on the board and saying: 'You discovered this pattern with your hands. Now let us write it in the language of mathematics.' Write step by step: 'Full circle has angle 360° and area πr^2. A sector with angle θ is $(\theta/360)$ of the full circle. Therefore, sector area $A = (\theta/360) \times \pi r^2$.' Underline $(\theta/360)$ and say: 'This fraction is the heartbeat of the formula — it tells us what share of the full circle the farmer has watered.' For Problem 1, do NOT solve it — say: 'Substitute the Laikipia values into the formula. You have 5 minutes. Work silently.' (WAIT TIME for productive struggle.) After 5 minutes, cold-call one learner to write their working on the board. Ask the class: 'Do you agree with each step? Thumbs up or thumbs down.' For any thumbs down, ask that learner: 'Which step do you disagree with, and what would you write instead?' (WAIT TIME 12 seconds.) Repeat for Problem 2. For the extension problem, tell early finishers: 'You are now the engineer — the area is given, find the angle. Rearrange the formula.' After Problems 1 and 2 are reviewed, ask the whole class: 'How does this formula help the Laikipia farmer plan a water budget?' Cold-call 2 learners. (WAIT TIME 10 seconds each.)	Worked Example → Guided Practice → Independent Practice (I Do – We Do – You Do) — the teacher models the derivation explicitly before releasing learners to apply it. Connecting each algebraic step back to the paper-circle investigation and the Laikipia phenomenon ensures conceptual coherence, not just procedural fluency. The extension problem builds 'Using Mathematics and Computational Thinking' (NGSS SEP 5) by requiring formula rearrangement.	Circulate during individual problem-solving. Observable indicators: (a) learner correctly substitutes $\theta = 120°$ and r = 40 m in Problem 1 to get $A = (120/360) \times \pi \times 40^2 = (1/3) \times 5026.55 \approx 1675.52$ m^2; (b) learner correctly gets $A = (75/360) \times \pi \times 6^2 \approx 23.56$ m^2 for Problem 2. Identify learners who substitute r instead of r^2 as a common misconception — address by asking: 'What is the formula for the full circle area? Now where does that appear in the sector formula?'

Driving Question Board (DQB) Creation 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or writes on a shared DQB section). On the first (green) sticky note, they write one thing they can now answer about the Laikipia farmer's problem. On the second (orange) sticky note, they write one new question that today's lesson has raised — something they still cannot calculate or are curious about (e.g., 'What if the watered region is not a full sector but a strange shape between two circles AND part of a rotation?'). Learners post both notes on the class Driving Question Board under the columns 'What we CAN now calculate' and 'What we STILL need to figure out.' The teacher reads 3–4 orange notes aloud and briefly previews that upcoming lessons will address the annular sector and segment.	Direct learners to the DQB (a large sheet of chart paper or a section of the board divided into columns). Say: 'In Lesson 1 you asked how to find the area of the ring between two pipes. Today you built a tool for the wedge the pipe sweeps. What new question does that raise?' Give 3 minutes for writing. After posting, read three orange notes aloud without revealing who wrote them. For each, say: 'This is a great question — hold onto it. We will need exactly this thinking in Lesson 3.' Point to the full driving question written at the top of the DQB and ask: 'Are we closer to answering this? Which part of the farmer's problem can we now solve, and which part is still open?' Cold-call 2 learners. (WAIT TIME 10 seconds each.)	Driving Question Board — a running metacognitive record that makes the trajectory of the learning sequence visible to learners. Distinguishing 'what we CAN calculate' from 'what we STILL need' mirrors the scientific practice of 'Asking Questions and Defining Problems' (NGSS SEP 1) and sustains curiosity across the 8-lesson sequence.	Read all green sticky notes after class. Observable indicator: learner correctly names the sector formula and links it to a specific part of the Laikipia scenario (e.g., 'I can now find the area of maize watered after 120° of rotation'). Learners whose green notes are vague ('I learned about sectors') receive a follow-up prompt card at the start of Lesson 3.
Model Building Phase 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open the circle-decomposition diagram they began in Lesson 1 (a large circle representing the full Laikipia field, with the annulus already labelled). Today they add a new layer: using a protractor, they draw a 120° sector inside the inner circle (r = 40 m) and label it with the formula $A = (\theta/360) \times \pi r^2$ and the calculated value ≈ 1675.52 m². They shade the sector in a distinct colour (e.g., yellow for 'watered maize'). Beneath the diagram, they write a one-sentence 'model statement': 'A sector is the region swept by the pivot pipe through angle θ, and its area is always the fraction θ/360 of the full circle.' Learners who finish early add a second sector (e.g., 240°) and calculate its area, then note how the two sectors together make up the whole circle.	Say: 'Your diagram is your growing map of the farmer's field. Every lesson, we add a new region. Today, add the sector — the wedge the pipe has swept. Use your protractor carefully; accuracy matters because the farmer is making real planting decisions.' Circulate and check that: (a) the sector is drawn from the centre (not from the edge); (b) the angle is marked with a protractor (not estimated); (c) the formula and numerical answer are both written. For any learner who draws the sector incorrectly (e.g., from the circumference), ask: 'Where does the pivot pipe start — at the edge of the field or at the water source in the centre?' (WAIT TIME 10 seconds.) Close the lesson by projecting a model diagram with all parts correctly labelled and say: 'By Lesson 8, your diagram will	Cumulative Model Revision — the learner's diagram serves as a persistent, evolving representation that tracks conceptual growth across the 8-lesson sequence. Adding each new region physically and labelling it with its formula builds a coherent visual argument for the driving question. This mirrors the NGSS Science and Engineering Practice of 'Developing and Using Models' (SEP 2).	Collect diagrams for a quick scan at the end of class. Observable indicators: (a) sector is drawn correctly from the centre with angle labelled; (b) formula $A = (\theta/360) \times \pi r^2$ is written; (c) the model statement sentence is present and accurate. Return diagrams with a brief written comment at the start of Lesson 3.

		show every region the farmer needs — rings, wedges, overlapping patches. Today you added the wedge. What do you think comes next?'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did learners' predictions in Phase 1 show evidence of connecting fractions to the full-circle area from Lesson 1, or did most learners treat this as a completely new problem? If few learners made the fraction connection independently, plan a brief bridging activity at the start of Lesson 3 to reinforce the link between fractional thinking and circle sub-regions. (2) During the paper-cutting investigation, which groups struggled most with using the protractor accurately? Were the errors random or systematic (e.g., always measuring from the wrong baseline)? Consider demonstrating protractor use again at the start of Lesson 3 before learners use it in the model-building phase. (3) Did the Kisumu roundabout context feel authentic to learners, or did it feel contrived? If learners engaged more with the Laikipia problem, lean further into the irrigation scenario for Lessons 3 and 4 and save urban planning contexts for the segment lesson. (4) How many learners successfully rearranged the formula for the extension problem (finding θ given area)? This algebraic manipulation will be needed again in Lesson 5 (segment) — note which learners need additional algebra support. (5) Were the orange DQB sticky notes focused on the next mathematical step (annular sector, segment) or on completely off-topic questions? If the latter, the phenomenon connection may need strengthening at the start of Lesson 3 by revisiting the aerial photograph and explicitly asking: 'We can now calculate the wedge. What other regions do we still see in this photograph that we cannot yet calculate?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners cut paper circles into sectors of 90°, 120°, 60°, and 45°, compared their areas as fractions of the whole circle, and completed a recording table showing how sector area changes with angle. They also examined an aerial-view sketch of the Laikipia centre-pivot field showing the 120° sweep of the pivot pipe.
What did I learn?	A sector is the 'pie-slice' region of a circle bounded by two radii and an arc. Its area is a fraction of the full circle's area: $A = (\theta/360) \times \pi r^2$. The Laikipia pivot pipe sweeping 120° waters approximately 1675.52 m^2 of maize (one third of the full field). The Kisumu roundabout pedestrian refuge sector ($r = 6$ m, $\theta = 75°$) has an area of approximately 23.56 m^2.
How does this explain the phenomenon?	Because a full circle spans 360°, any sector with central angle θ covers exactly θ/360 of the total area. Multiplying this fraction by the full-circle area (πr^2) gives the sector area formula $A = (\theta/360) \times \pi r^2$. This allows the Laikipia farmer to calculate precisely how much of the maize field has been watered at any point in the pivot's rotation, enabling accurate water budgeting and planting decisions.

LESSON 3 (80 minutes): Arc Length and Sector Perimeter: Fencing the School Garden in Nakuru	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive and apply the arc length formula l = (θ/360) × 2πr and calculate the full perimeter of a sector, contextualised around fencing a sector-shaped school garden in Nakuru, as a foundation for computing segment area in later lessons.
Knowledge	By the end of the lesson, learners should be able to: (a) state that an arc is a fraction of the full circumference proportional to the angle at the centre; (b) derive the arc length formula l = (θ/360) × 2πr; (c) recall that the perimeter of a sector = arc length + 2 radii (P = l + 2r); (d) connect arc length to the anchoring phenomenon of the pivot-irrigation pipe that has swept a partial angle.
Skills	By the end of the lesson, learners should be able to: (a) calculate arc length given radius and central angle in degrees; (b) calculate the full perimeter of a sector; (c) select and substitute correct values into l = (θ/360) × 2πr; (d) estimate reasonableness of answers by comparing arc length to full circumference.
Attitudes	Learners will: (a) appreciate that precise measurement prevents material waste and saves money for farmers and artisans; (b) show confidence in deriving formulae from first principles rather than memorising; (c) value collaborative reasoning when checking each other's calculations.
Key Inquiry Question	How do we determine the arc length and perimeter of a sector, and how are these applied in real-life situations such as fencing and irrigation planning?
Purpose in Storyline	In Lesson 1 learners mastered the annulus area (the ring between the two pivot pipes). In Lesson 2 they derived the sector area formula A = (θ/360) × πr². Now in Lesson 3 they discover that planning the farmer's fence or a school garden boundary also requires the arc length — the curved edge of that wedge. Without arc length, a fencing contractor cannot order the right amount of wire, and without the full sector perimeter the garden designer cannot quote a price. This lesson therefore builds the last missing measurement tool before Lesson 4 tackles segment area, where both arc length and sector area will be combined.
Safety Notes	No hazardous materials are used. If learners use compasses for drawing, remind them to keep the sharp point directed downward onto paper and not toward classmates. Ensure adequate desk space so that metre rulers and string do not sweep into adjacent learners.

B. LESSON OVERVIEW

Learners begin by predicting how much fencing wire is needed to enclose a sector-shaped school garden in Nakuru, given only the radius and central angle. Through a string-and-circle physical activity they discover that an arc is a proportional fraction of the full circumference. They then formalise the arc length formula l = (θ/360) × 2πr, extend it to the full sector perimeter P = l + 2r, and solve graded problems set in Kenyan contexts (irrigation pivots in Laikipia, curved market stalls in Kisumu, a Nakuru school garden). The lesson closes with learners updating their Driving Question Board and revising their cumulative circle-decomposition models to include arc length as a labelled edge.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a scaled	Display the sector diagram on the	Anchored Prediction with Public	Teacher scans exercise books

📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chords in Circles: Arc and Chord (PLIX) Source: CK-12 🔍 Search ARES for similar readings	diagram of a sector-shaped school garden at Nakuru High School: radius 14 m, central angle 90°. The prompt is displayed: 'The school wants to fence all three sides of this garden. Estimate the total length of fencing wire needed. Write your estimate and your reasoning in your exercise book — do not calculate yet, just reason.' Learners work individually for 3 minutes, then share with a shoulder partner for 2 minutes. Selected pairs share with the class. Learners also reconnect to the phenomenon: 'Think back to the Laikipia pivot farm — if the irrigation pipe has swept a 90° angle and the farmer wants to fence that watered wedge, what length of curved wire would the outer edge need?'	board (or draw it clearly with chalk: r = 14 m, θ = 90°). Say exactly: 'Before we do any calculation, I want your best reasoned guess. Think about what you already know about circles and circumference.' Allow 3 minutes of silent individual thinking — do not intervene. After 3 minutes say: 'Turn to your shoulder partner and compare your estimates. Do you agree? Why or why not?' Allow 2 minutes. Cold-call 3 pairs (choose one high-estimate pair, one low-estimate pair, one mid-range pair). For each, ask: 'What reasoning did you use to arrive at that number?' Record all three estimates visibly on the board under the heading 'Our Predictions'. Do NOT confirm or correct at this stage. Then ask the reconnect question: 'Who can link this garden fence problem to what the Laikipia farmer needs to know?' Wait 10 seconds. Cold-call 2 learners.	Recording — learners commit to a visible estimate before instruction, creating cognitive dissonance that motivates the derivation. Recording all estimates publicly ensures every learner has a stake in discovering who was closest and why.	during partner talk to check whether learners are using circumference reasoning ($C = 2\pi r$) or simply guessing randomly. Observable indicator: at least 60% of learners reference circumference or fractions of a circle in their written reasoning. If fewer do, teacher asks a bridging question: 'What is the full distance around a circle of radius 14 m?' before proceeding.
Observe Phase 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chords in Circles: Arc and Chord (PLIX) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — String and Circle Exploration (Groups of 4, 12 minutes): Each group receives a cardboard circle (cut from a cereal box or manila paper) with radius 7 cm, a piece of string, a protractor, and a ruler. Step 1: Learners measure and record the full circumference by wrapping string around the circle, then measuring the string ($C = 2\pi r$ check). Step 2: Teacher instructs groups to mark a 90° sector on their circle using the protractor. Learners cut a piece of string to match exactly the curved arc of that sector. They measure the string length. Step 3: Learners repeat for a 180° sector and a 60° sector on the same circle. Step 4: Groups complete a results table: | Angle θ | Arc string length (cm) |	Distribute materials before the lesson begins to save time. As groups work on Activity A, circulate and ask targeted questions: 'What fraction of the full circle is a 90° sector?' (Wait 10 seconds.) 'How does your string length compare to the full circumference string?' After groups complete their tables, cold-call one group to read their four rows aloud. Ask the class: 'Does every group see the same pattern?' Wait 10 seconds for hands. If a group has inconsistent data, ask: 'What might have caused that measurement to differ?' For Activity B, do not simply write the formula — narrate the logic: 'We are asking: what fraction of the whole circumference does this arc represent? The	Concrete-to-Abstract Bridging via Measurement — learners physically measure arcs with string before seeing the algebraic formula. The results table makes the proportional relationship visible as a pattern across three cases, so the formula emerges from evidence rather than authority. This mirrors NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data) applied in a mathematical context.	Check groups' results tables as you circulate. Observable indicator: all four groups correctly compute θ/360 to 2 decimal places for at least two of the three angles and observe it matches Arc/Circumference. During Activity B, the thumbs check after each derivation step gives real-time data. If more than 4 learners show thumbs sideways after the fraction step, pause and re-explain using the analogy: 'If I cut a mandazi (doughnut) into 4 equal pieces and take one piece, I have taken 1/4 = 90/360 of the whole.'

	Arc ÷ Circumference | θ ÷ 360 |. Learners observe that (Arc ÷ Circumference) equals (θ ÷ 360) in every row. ACTIVITY B — Worked Example Reveal (5 minutes): Teacher reveals the full worked derivation on the board: 'Arc length is a fraction of the full circumference. The fraction is θ/360. So l = (θ/360) × 2πr.' Teacher then shows the Nakuru garden example step-by-step: r = 14 m, θ = 90°; l = (90/360) × 2 × π × 14 = (1/4) × 88 = 22 m. Perimeter of sector = l + 2r = 22 + 28 = 50 m. Learners copy and annotate in their books.	whole circle is 360°, our sector is θ degrees, so the fraction is θ over 360. We multiply that fraction by the full circumference 2πr.' Pause after each line and ask: 'Does this step make sense? Thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-sideways or thumbs-down immediately.		
Explain Phase 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chords in Circles: Arc and Chord (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — Guided Practice (10 minutes): Learners solve three graded problems in their exercise books, working individually. Problem 1 (Basic): An irrigation pivot in Laikipia has swept through 120°. The pipe is 21 m long. Calculate the arc length traced by the tip of the pipe. Problem 2 (Intermediate): A curved market stall in Kisumu Night Market is shaped like a sector with radius 3.5 m and central angle 150°. Calculate the total length of fencing needed to enclose the stall. Problem 3 (Extended): A sector-shaped maize plot at the Laikipia farm has radius 28 m and central angle 72°. The farmer wants to plant sunflowers along the curved boundary only. Calculate the arc length. Then calculate the total perimeter if the farmer also fences the two straight sides. PART B — Peer Marking and Error Discussion (5 minutes): Learners swap books with their shoulder partner and mark using the teacher's displayed solutions. Partners circle any step where	Before releasing learners to individual work, say: 'You have 10 minutes. Attempt all three. If you finish early, write one sentence explaining how arc length connects to the Laikipia farmer's fencing problem.' Circulate and observe: are learners writing the formula first, then substituting? Are they using correct order of operations? Do not give answers — instead ask: 'What is the first thing you need to identify before substituting into the formula?' (Wait 10 seconds.) After 10 minutes, display full worked solutions on the board. For the peer-marking phase, say: 'If your partner has a different answer to yours, your job is not to say who is wrong — your job is to find the first step where you made different choices.' Cold-call 2 pairs to share a disagreement they found. For the common error discussion, write the wrong working on the board (e.g., P = l only, omitting 2r) and ask: 'What has this learner forgotten about the shape of a sector?' Wait 15 seconds. Cold-	Graduated Release with Peer Accountability — the three problems increase in demand (arc only → full perimeter → context-rich extension), ensuring every learner experiences success at level 1 while being stretched at levels 2 and 3. Peer marking transfers explanatory responsibility to learners, deepening understanding through articulation of errors. This reflects NGSS Practice 6 (Constructing Explanations).	Collect 6 exercise books (2 strong, 2 mid, 2 developing learners) to review after the lesson for misconceptions. During circulation, track on a clipboard: (a) How many learners correctly set up the formula before substituting? (b) How many correctly add 2r for Problems 2 and 3? Observable success indicator: ≥75% of learners correctly solve Problem 1; ≥60% correctly solve Problem 2 (full perimeter); ≥40% correctly solve Problem 3 in the time allowed.

	they disagree and discuss the discrepancy. Teacher selects one common error (likely: forgetting to add 2r for the full perimeter, or using diameter instead of radius) and addresses it with the whole class.	call 3 learners.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Chords in Circles: Arc and Chord (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front. The board already has question cards from Lessons 1 and 2 (annulus area, sector area). Each learner receives a sticky note. They write: (a) one thing today's lesson now allows them to answer about the Laikipia farmer's problem (green note), OR (b) one new question today's lesson has raised (orange note). Learners post their notes and the class does a 3-minute gallery walk to read peers' notes. Teacher leads a 3-minute whole-class discussion to cluster the new questions (e.g., 'Can we use arc length to find the area of the curved segment between the arc and a chord?' — this previews Lesson 4). The teacher adds a new header card to the DQB: 'Arc Length & Sector Perimeter — SOLVED ✓' and moves the open segment question to the 'Still investigating' column.	Hand out sticky notes as learners are finishing Problem 3. Say: 'Think about our Laikipia farmer and the Nakuru school garden. Write one thing you can now calculate that you could not calculate at the start of today's lesson, or one new question today has raised for you.' Allow 3 minutes for writing. During the gallery walk, circulate and read notes — identify 2 green notes and 2 orange notes to highlight in discussion. Start discussion: 'I noticed several of you wrote about the curved region between the arc and a straight line across the sector — what is that shape called?' Wait 10 seconds. Accept responses (segment). 'That is exactly what Lesson 4 will tackle. What do you think we will need from today's lesson to find the area of that segment?' Wait 15 seconds. Cold-call 3 learners. Update the DQB visibly so learners see their growing understanding mapped spatially.	Public Knowledge Mapping — the DQB makes the cumulative structure of the unit visible. Moving cards from 'investigating' to 'solved' gives learners a tangible sense of progress and helps them see how arc length (today) will feed into segment area (Lesson 4). This supports NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information).	Read all sticky notes after the lesson. Count: (a) How many green notes correctly name something arc length or sector perimeter enables? (b) Do any orange notes reveal a persistent misconception (e.g., confusing arc length with sector area)? This informs the opening of Lesson 4. Observable indicator: at least 70% of green notes correctly reference arc length, fencing, or perimeter in a meaningful sentence.
Model Building Phase 📹 **VIDEO:** Circumference Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Learners open their cumulative circle-decomposition diagrams begun in Lesson 1 (a large labelled circle diagram showing the full circle, annulus, and sector). Today they add two new annotations: (1) On the curved edge of the sector they have already drawn, they write the arc length formula $l = (\theta/360) \times 2\pi r$ and calculate and label the arc	Say: 'Take out your running Laikipia circle diagram. We are going to update it with what we learned today.' Circulate and check that learners are annotating the curved edge (not a radius). Prompt: 'Where exactly on your diagram does the arc live? Point to it before you write the formula.' For learners who are adding the chord extension: 'If you draw a straight	Cumulative Model Revision — learners add to the same diagram across all 8 lessons, making the decomposition of the full circle into sub-regions progressively richer and more precise. Each lesson's addition is visible on the same page, so the model literally grows as understanding grows. This reflects NGSS Practice 2 (Developing and Using Models)	Visual scan of held-up diagrams gives immediate whole-class data. Specific observable indicators: (a) arc length formula is written on the curved edge, not on a radius; (b) sector perimeter bracket clearly shows $P = l + 2r$; (c) numerical values are correctly calculated for r = 21 m, θ = 90° (l = 33 m, P = 75 m). The exit task (r = 7 m, θ = 60°) is collected at the start of Lesson 4

Chords in Circles: Arc and Chord (PLIX) Source: CK-12 🔍 Search ARES for similar readings	length for the specific values used in their diagram (r = 21 m, θ = 90°, from the running Laikipia scenario). (2) They draw a bracket showing the full sector perimeter P = l + 2r and record its value. At the bottom of the diagram, learners write a one-sentence 'model statement': 'The arc is a fraction θ/360 of the full circumference, and the sector perimeter includes both the arc and the two radii.' Learners who finish early begin sketching and labelling where the chord of a segment would appear — setting up Lesson 4.	line connecting the two ends of the arc, what region does that create between the chord and the arc?' (Wait 10 seconds — do not answer; this is their preview for Lesson 4.) In the final 2 minutes, ask learners to hold up their diagrams. Teacher does a quick visual scan: is the arc labelled on the curved edge? Is the perimeter bracket visible? Address any common errors whole-class before dismissal. Assign exit task: 'Calculate the arc length and perimeter of a sector with r = 7 m and θ = 60°. Show all working. Bring it tomorrow.'	and ensures continuity across the evidence-gathering sequence.	to inform grouping and targeted support.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did the string-and-circle activity (Activity A) generate the proportional pattern clearly enough for most learners to articulate the formula themselves, or did you need to guide more heavily than expected? (2) In the guided practice, which of the three problems caused the most errors — was it the formula setup, the arithmetic, or the addition of 2r for the perimeter? Adjust the opening of Lesson 4 accordingly. (3) Were the DQB sticky notes revealing genuine conceptual growth, or were learners copying the formula without connecting it to the phenomenon? If the latter, plan a brief 5-minute reconnect at the start of Lesson 4 asking: 'What does the arc length tell the Laikipia farmer that sector area alone cannot?' (4) Did learners with lower numeracy confidence participate meaningfully in the string activity? If not, consider assigning them the role of 'recorder' in the results table in future group activities so they are actively involved in the data-generation phase. (5) Review the 6 collected exercise books for the specific error pattern (diameter vs. radius, or omitting 2r) and prepare one targeted worked example to open Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners wrapped string around cardboard circle arcs of three different central angles (90°, 180°, 60°) and measured each arc length, recording results in a table that compared Arc÷Circumference with θ÷360, discovering the proportional relationship directly from physical measurement.
What did I learn?	The arc length of a sector is a fraction of the full circumference equal to the ratio of the central angle to 360°, giving the formula $l = (\theta/360) \times 2\pi r$; the full perimeter of a sector is this arc length plus the two radii, $P = l + 2r$.
How does this explain the phenomenon?	Learners applied the arc length and sector perimeter formulae to three graded Kenyan contexts — an irrigation pivot in Laikipia, a curved market stall in Kisumu, and a sector-shaped maize plot — and updated their cumulative circle-decomposition diagrams to show arc length as a labelled curved boundary, connecting the mathematics directly to the fencing and planning needs of the anchoring phenomenon farmer.

LESSON 4 (80 minutes): Area of an Annular Sector: Combining the Ring and the Wedge	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive and apply the annular sector formula $A = (\theta/360) \times \pi(R^2 - r^2)$ by combining their prior understanding of annulus area and sector area, enabling them to calculate the exact area swept by the outer irrigation ring in the Laikipia centre-pivot system and to solve the Kisumu road-marking problem.
Knowledge	By the end of the lesson, learners should be able to: (a) state the formula for the area of an annular sector as $A = (\theta/360) \times \pi(R^2 - r^2)$; (b) explain how the annular sector formula is derived by combining the annulus formula and the sector formula; (c) identify annular sectors in real-life contexts including irrigation rings and road markings.
Skills	By the end of the lesson, learners should be able to: (a) derive the annular sector formula systematically from first principles using the sector and annulus formulae; (b) substitute correctly into $A = (\theta/360) \times \pi(R^2 - r^2)$ to calculate areas; (c) apply the formula to at least two real-life Kenyan contexts (Laikipia irrigation and Kisumu road markings).
Attitudes	Learners appreciate that mathematical formulae are not isolated rules but are built by combining simpler ideas, and that accurate calculation prevents waste of resources such as water, seed, and paint in Kenyan agriculture and infrastructure.
Key Inquiry Question	How do we determine the area of parts of a circle, and how are areas of parts of a circle applied in real-life situations?
Purpose in Storyline	In Lessons 1–3 the class derived formulae for the full annulus (ring) and the sector (wedge). The Laikipia farmer's problem now requires both ideas at once: the outer irrigation pipe sweeps only part-way around the field, creating a wedge-shaped ring — an annular sector. Without this formula the farmer cannot calculate how much of the outer ring has been watered at any given angle, making water budgeting and seed planning impossible. Lesson 4 closes that gap.
Safety Notes	No hazardous materials are used. Compasses have sharp points — learners must place the compass on a firm surface and must not point compasses at other learners. Remind learners to handle geometrical sets carefully.

B. LESSON OVERVIEW

Learners open by predicting the area formula for a wedge-shaped ring (annular sector) before any instruction. They then observe and manipulate a paper cut-out activity and a worked diagram, gathering evidence that the annular sector is simply the difference between two sectors of the same angle but different radii. Through structured sensemaking, groups derive $A = (\theta/360) \times \pi(R^2 - r^2)$ and verify it against their predictions. The formula is applied to (i) the Laikipia outer irrigation ring swept through a given angle and (ii) a curved road-marking strip in Kisumu. Learners update the class Driving Question Board and revise their cumulative circle-decomposition model to include the new sub-region. Formative assessment occurs through cold-call questioning, mini-whiteboard checks, and a two-question exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** The future of	Learners receive a printed aerial-photograph sketch of the Laikipia farm showing two concentric	Display the sketch on the board (or distribute printed copies). Say: 'Look carefully at the shaded	Prediction-with-reasoning (Eliciting Prior Knowledge): By committing a prediction to paper before	Circulate during the 4-minute individual prediction phase. Look for: (a) learners who attempt to

creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	circles (inner radius r = 40 m, outer radius R = 70 m) with a sector angle of 120° shaded between them. Working individually for 4 minutes, each learner writes a prediction: 'I think the area of the shaded region is ___ m² because ___.' They are explicitly told NOT to look at previous notes — this is a raw prediction. Learners then share predictions with a shoulder partner for 2 minutes, noting where they agree or disagree.	region on the Laikipia farm map. This is the part of the outer irrigation ring that has been watered so far. In your exercise books, write your best prediction for its area and your reasoning. You have 4 minutes — work in silence.' Set a visible timer. After 4 minutes say: 'Turn to your neighbour. Compare your predictions. Do NOT change your answer yet — just note if you agree or disagree.' After 2 minutes of pair talk, cold-call 3 learners: 'Achieng, what did you predict and why?', 'Mwangi, did your partner agree with you?', 'Fatuma, which formula from our previous lessons did you try to use?' Allow 10 seconds of WAIT TIME after each question before accepting responses. Record 3–4 different predictions on the board without evaluation — label them P1, P2, P3.	instruction, learners activate their existing schema for annulus area (Lesson 1) and sector area (Lesson 2). Recording multiple predictions on the board creates a public record that motivates the subsequent investigation — learners are genuinely curious about which prediction is closest.	subtract two sector areas — evidence of strong prior integration; (b) learners who write only πR^2 or only $\pi(R^2-r^2)$ — evidence they have not yet connected sector and annulus ideas; (c) learners who leave the reasoning blank — seat these learners near a peer who will be asked to share. Do not correct errors at this stage; note which misconceptions appear for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 receive an evidence-gathering envelope containing: (1) two pre-cut paper sectors of the SAME angle θ = 90° — one large sector (radius R = 14 cm) and one small sector (radius r = 7 cm), both made from grid paper so area can be counted; (2) a ruler and scissors. Task A (5 min): Each group places the small sector exactly over the large sector, aligns the vertices and one straight edge, then physically removes the small sector from the large sector. They observe the leftover shape and sketch it, labelling it 'annular sector.' Task B (5 min): Using the grid squares, groups count (estimate) the area of the leftover annular sector by counting full and half squares. Task C (5 min): Groups calculate $(\theta/360)\times\pi R^2$ and	Distribute envelopes. Say: 'You have four tasks — the instructions are on your worksheet. Your goal is to discover what the area of an annular sector equals before I tell you. Use the grid paper, your rulers, and what you already know about sectors and annuli.' Circulate continuously. At the 5-minute mark prompt: 'Has every group placed the small sector over the large one and removed it? What shape is left? Give it a name.' At 10 minutes prompt Task C: 'Now use the formulae you already know. Calculate the area of the large sector alone. Then the small sector alone. Write both down.' At 13 minutes: 'Subtract the small sector area from the large sector area. Compare that number to your counted grid squares.	Concrete Manipulation followed by Numerical Verification (Evidence-to-Pattern): Physically removing a small sector from a large sector makes the subtraction structure of the formula tangible before it is symbolic. Counting grid squares provides an independent check that grounds abstract calculation in observable evidence, supporting NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data).	Check each group's results table. Success indicator: the calculated difference $(\theta/360)\times\pi(R^2-r^2)$ is within 5% of the grid-count estimate. Common error to watch for: learners subtract radii first $(R-r)$ and then square — i.e., they write $(\theta/360)\times\pi(R-r)^2$ instead of $(\theta/360)\times\pi(R^2-r^2)$. If this error appears in 2 or more groups, flag it for direct address in Phase 3 without naming the group.

	$(\theta/360)\times\pi r^2$ separately using their known radii ($R=14$ cm, $r=7$ cm, $\theta=90°$), then subtract. They record both the counted area and the calculated area in a results table. Task D (3 min): Groups examine the road-marking diagram on the reverse of their worksheet — a curved white road-marking strip in Kisumu's Mega roundabout ($R = 12$ m, $r = 9$ m, $\theta = 60°$) — and record the given measurements without yet calculating.	What do you notice?' Cold-call 2 groups to share their results table: 'Nyambura's group — what did you count, and what did your formula give?' Allow 10 seconds WAIT TIME. Write both values on the board next to the group name. Say: 'Every group whose counted area and calculated area are within 2 cm^2 of each other — raise your hand.' Affirm the pattern without yet stating the formula.		
Explain Phase 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class guided derivation (12 minutes): The teacher leads learners to write the derivation formally in their books step by step. Learners then work in pairs on two application problems: Problem 1 (Laikipia): The outer irrigation ring has $R = 70$ m, $r = 40$ m, and the pivot arm has rotated 120°. Calculate the area of the outer ring that has been watered. Problem 2 (Kisumu road marking): A curved road-marking strip at Kisumu's Mega roundabout has $R = 12$ m, $r = 9$ m, $\theta = 60°$. Calculate the area of paint needed (to the nearest 0.1 m^2). Pairs write full working — formula, substitution, evaluation — and box their final answers with units. After 10 minutes of pair work, pairs compare answers with the neighbouring pair (group of 4) and reconcile any differences. Each learner then individually writes a one-sentence explanation: 'The annular sector formula works because ___.'	Begin derivation by pointing to the board predictions from Phase 1: 'Let us see which prediction is correct. We will build the formula from what we already know.' Write on the board: 'Area of large sector $= (\theta/360) \times \pi R^2$' and 'Area of small sector $= (\theta/360) \times \pi r^2$.' Ask: 'If I want only the region BETWEEN the two sectors — what operation do I perform?' Wait 10 seconds. Cold-call 3 learners by name. Accept subtraction. Write: 'Area of annular sector $= (\theta/360)\times\pi R^2 - (\theta/360)\times\pi r^2$.' Ask: 'What is the common factor in both terms?' Wait 10 seconds. Cold-call 2 learners. Factor out: '$= (\theta/360) \times \pi(R^2 - r^2)$.' Circle this result in coloured chalk and say: 'This is our formula for today. Write it in your formula box and underline it.' Address the $(R-r)^2$ misconception directly: draw both expressions on the board, substitute $R=14$, $r=7$ and show the numerical difference. Say: 'Squaring each radius FIRST and then subtracting gives a different — and correct — answer from subtracting first and then squaring. The order matters.' For Problem 1, say: 'You have 10 minutes in pairs. Show every step. A correct answer without working	Algebraic Factoring as Conceptual Bridge (Connecting Representations): Factoring $(\theta/360)$ out of both terms shows learners that the annular sector formula is not a new isolated rule but an algebraic consequence of two formulae they already know. This deepens conceptual understanding (NGSS SEP 6 — Constructing Explanations) and reduces cognitive load by anchoring new knowledge to existing schema. The one-sentence explanation consolidates understanding through language production.	Circulate during pair work. Collect 4–6 exercise books and scan the working for: (a) correct substitution of R^2 and r^2 (not $(R-r)^2$); (b) correct unit (m^2); (c) correct factoring step shown. If more than one-third of pairs make the $(R-r)^2$ error during Problem 1, pause the class and re-teach the substitution step using a different numerical example ($R=10$, $r=6$, $\theta=180°$) before releasing them to Problem 2. Use mini-whiteboards or slates: ask each learner to hold up their final answer to Problem 1 simultaneously — scan for outliers.

		earns zero marks in KNEC.' Circulate; when pairs finish, prompt: 'Did you include units? Are your units m²?' After pair-compare, cold-call 2 pairs to write their working on the board. Ask the class: 'Do you agree or disagree? Thumbs up or down.' Resolve disagreements through student explanation, not teacher correction where possible.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives one sticky note (or tears a small piece of paper). They write ONE thing the Laikipia farmer can now calculate that they could not calculate at the start of today's lesson. Learners post their notes on the class DQB under the column 'What we CAN now explain.' The teacher reads aloud 4–5 notes and the class clusters them into themes. Learners also scan the 'Questions we still have' column from previous lessons and identify any question that Lesson 4 has now answered — these are moved to 'Answered.' Finally, learners add any NEW questions that arose during today's lesson to the 'Questions we still have' column (e.g., 'What if the two circles are not concentric?' or 'How do we find the area where two irrigation circles OVERLAP?').	Say: 'Think about the Laikipia farmer and the problem we started with. On your sticky note write ONE specific thing the farmer can now calculate thanks to today's formula. Be precise — name the shape and the formula.' Give 2 minutes of silent writing. As learners post notes, read 4–5 aloud and ask: 'Does anyone want to add to this idea, or does anyone see a note that says something different?' Point to the 'Questions we still have' column: 'Look at questions posted in Lessons 1, 2, and 3. Does anyone see a question that today's lesson has answered?' Cold-call 2 learners. Then ask: 'Did today's lesson open any NEW questions — things you are now curious about that we have not answered yet?' Accept 2–3 new questions and post them. Close by reading the driving question aloud: 'How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?' Ask: 'Show me on your fingers: 1 = we can barely answer this, 5 = we can answer it fully. Hold up your fingers.' Survey the room and note the distribution.	Public Knowledge Tracking (DQB Protocol): The DQB makes the class's collective epistemic progress visible and continuous. Moving cards from 'Questions we still have' to 'Answered' gives learners a tangible sense of growing explanatory power. New questions posted by learners model the scientific practice that answers generate further questions (NGSS SEP 1 — Asking Questions), sustaining curiosity across the 8-lesson sequence.	Read the sticky notes before the next lesson. Success indicator: learners refer specifically to the annular sector (not just 'a new formula') and connect it to a concrete quantity (area of outer ring watered, litres of water used, seeds needed). Vague notes ('we learned a new formula') indicate shallow encoding — these learners should be seated at the front for Lesson 5 and given sentence-starter scaffolds.
Model	Learners retrieve their cumulative	Say: 'Take out your Circle	Cumulative Annotated Modelling	Mark exit tickets before Lesson 5.

Building Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Circle Decomposition Model' — a large annotated circle diagram that has been updated across Lessons 1–3 to show the full circle, annulus, and sector. Today each learner adds a new shaded region to their model: the annular sector. They label it with: (a) its name in English and in Kiswahili (sekta ya pete); (b) the formula $A = (\theta/360) \times \pi(R^2 - r^2)$; (c) a small real-life Kenyan example (Laikipia irrigation ring or Kisumu road stripe). They also add an arrow and label showing which previous sub-regions (annulus + sector) combine to produce the annular sector, making the derivation path visible in the model. Learners write a two-sentence 'model update note' at the bottom of their diagram: 'Before Lesson 4, my model could show ___. After Lesson 4, my model can also show ___.' They share models with a partner for 2 minutes of peer feedback before the exit ticket.	Decomposition Model from your portfolio. You are going to add the annular sector as a new region. Use a different colour for this lesson's addition so the growth of your model is visible over time.' Give 7 minutes. Circulate and look for: correct formula written in the label; an arrow connecting 'annulus' and 'sector' to 'annular sector' showing the derivation logic; a Kenyan real-life example. Prompt learners who only write the formula: 'Can you draw an arrow from the annulus label and the sector label both pointing TO the annular sector label? That arrow shows HOW we derived it.' After 7 minutes say: 'Show your model to your partner. Your partner should check: is the formula correct? Is the derivation arrow there? Is there a Kenyan example?' Give 2 minutes of peer feedback. Then say: 'In the last 3 minutes, complete your exit ticket silently and independently.' Distribute the exit ticket: Q1 — A curved path at Uhuru Gardens, Nairobi, forms an annular sector with R = 8 m, r = 5 m, and θ = 150°. Calculate its area to 2 decimal places. Q2 — A farmer in Laikipia extends the pivot arm so R increases from 70 m to 90 m. The inner radius stays r = 40 m and the angle stays 120°. By how many m^2 does the watered annular sector area increase? Collect exit tickets as learners leave.	(Progressive Model Revision — NGSS SEP 2): Each lesson's model update makes the learner's growing understanding of circle decomposition externally visible and revisable. Derivation arrows between sub-regions communicate that mathematical knowledge is structured and connected, not a collection of isolated formulae. The 'before/after' model update note promotes metacognitive reflection on conceptual change.	Success threshold: Q1 correct to 2 d.p. ($A = (150/360)\times\pi\times(64-25) = (5/12)\times\pi\times39 \approx 51.05$ m^2) and Q2 shows correct subtraction of two annular sector areas ($(120/360)\times\pi\times(8100-1600) - (120/360)\times\pi\times(4900-1600) = (1/3)\times\pi\times(6500-3300) = (1/3)\times\pi\times3200 \approx 3351.03$ m^2). Learners who answer Q1 correctly but fail Q2 likely cannot yet extend the formula to a comparison context — group these learners for a targeted 5-minute review at the start of Lesson 5. Learners who fail Q1 need immediate intervention on substitution order (R^2 first, then r^2).

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal: (1) How many learners made the $(R-r)^2$ substitution error during Phase 2 or Phase 3? Was the direct-address strategy in Phase 3 sufficient to resolve it, or does it need revisiting at the start of Lesson 5? (2) Did the paper cut-out activity in Phase 2 make the subtraction structure visible to ALL learners, or did some groups skip to formula use without engaging with the physical manipulation? If so, consider replacing the cut-out with a GeoGebra dynamic diagram for Lesson 5. (3) Review the DQB sticky notes: do learners' new questions point toward the overlapping-circles context (the next lesson cluster), or are there gaps in understanding of annular sector itself that must be addressed first? (4) Were the two exit-ticket problems appropriately differentiated? Q2 required two calculations and a subtraction — note how many learners attempted it versus how many left it blank. Adjust Lesson 5's opening review accordingly. (5) Did the Kisumu road-marking context feel authentic to learners, or did it need more visual support (e.g., a photograph of a Kenyan roundabout road marking)? Source a photograph for future use.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners physically removed a small paper sector from a large paper sector of the same angle, producing a leftover annular-sector shape. They counted grid squares to estimate its area, then calculated $(\theta/360)\times\pi R^2$ and $(\theta/360)\times\pi r^2$ separately and subtracted — finding that the formula and the grid count agreed closely. They also examined measurements of a curved road-marking strip at Kisumu's Mega roundabout.
What did I learn?	The area of an annular sector — the wedge-shaped ring swept by the outer irrigation pipe in Laikipia — is found by subtracting the area of the inner sector from the area of the outer sector: $A = (\theta/360) \times \pi(R^2 - r^2)$. This formula combines the annulus formula from Lesson 1 and the sector formula from Lesson 2. Importantly, each radius must be squared BEFORE subtracting — not the other way around.
How does this explain the phenomenon?	Using the annular sector formula $A = (\theta/360) \times \pi(R^2 - r^2)$, the Laikipia farmer can calculate exactly how much of the outer irrigation ring has been watered at any angle, enabling accurate water budgeting, seed planning, and fencing estimates. The same formula solves the Kisumu road-marking problem, showing that one mathematical tool applies across agriculture and urban infrastructure.

LESSON 5 (80 minutes): Area of a Segment of a Circle	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive and apply the formula for the area of a segment of a circle: $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$, by reasoning that a segment is what remains when a triangle is subtracted from a sector.
Knowledge	By the end of the lesson, learners should be able to: (a) state that a segment is the region between a chord and its arc; (b) derive the segment area formula $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ by subtracting the area of the triangle ($\frac{1}{2}r^2\sin\theta$) from the area of the sector ($(\theta/360)\pi r^2$); (c) calculate the area of a segment given the radius and central angle.
Skills	By the end of the lesson, learners should be able to: (a) decompose a circle diagram into a sector and a triangle; (b) apply the formula $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ to solve real-life problems including the Turkana borehole cover hatch and a chord-cut region on a sufuria lid; (c) select the correct sub-formula depending on whether the segment is minor or major.
Attitudes	Learners appreciate how precise geometric decomposition — breaking a complex shape into manageable parts — enables Kenyan farmers, engineers, and artisans to avoid material waste and make accurate plans.
Key Inquiry Question	How do we determine the area of parts of a circle?
Purpose in Storyline	In Lesson 4, learners derived the annular sector formula and could calculate the watered ring-wedge on the Laikipia farm. However, when the pivot pipe does not swing a full wedge-shaped slice but instead the relevant boundary is a straight chord (e.g., a fence line cuts across the circular plot, or a flat hatch cover sits across a circular borehole), the sector formula alone over-counts by including the triangular region that is not part of the curved patch. In Lesson 5, learners subtract that triangle to isolate the true segment area, giving the farmer — and a Turkana water-project engineer — the exact area of the chord-cut region.
Safety Notes	No hazardous materials are used. If physical circular objects (sufuria lids, borehole cover cut-outs) are brought in, ensure edges are smooth or taped. Remind learners to handle calculators and geometrical sets carefully.

B. LESSON OVERVIEW
Learners begin by predicting how a chord divides a sector diagram of the Laikipia pivot field. Through a guided paper-folding and cut-out activity using circular discs, they physically separate a sector into a triangle and a curved segment, then measure and compare areas. Using their prior knowledge of sector area ($(\theta/360)\pi r^2$) and the triangle area formula ($\frac{1}{2}r^2\sin\theta$) from Lesson 4 trigonometry connections, learners derive $A_{segment} = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. They then apply this formula to two Kenyan real-life contexts: (1) calculating the area of a circular hatch cover on a Turkana borehole that is cut by a straight locking bar (chord), and (2) finding the area of the curved region remaining when a straight cut is made across a sufuria lid of known diameter. The lesson closes with learners updating the Driving Question Board and revising their cumulative circle-decomposition model to include the segment.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are shown a projected aerial sketch of the Laikipia pivot	Display the annotated Laikipia farm diagram and say: 'The farmer	Prediction with annotated sketch (Predict-Observe-Explain launch).	Circulate and scan sketches. Look for: (a) learners who correctly

📹 **VIDEO:** [Proof: radius is perpendicular to a chord it bisects] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Midpoints and Segment Bisectors (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	farm from Lesson 1. A new overlay is added: a straight fence line (chord) cuts across one sector of the watered field, creating a curved region on one side. Learners are asked to individually sketch what they think the 'curved patch' between the chord and the arc looks like, estimate whether its area is more or less than half the sector area, and write a prediction sentence: 'I think the area of the curved patch is ______ because ______.' They also predict whether the triangle left on the other side of the chord has a larger or smaller area than the curved patch.	has just put up a straight fence across one sector of the pivot field — look at the chord drawn here. Before we calculate anything, I want you to sketch the two regions this chord creates inside the sector and write your prediction.' Allow 3 minutes of silent individual thinking (WAIT TIME: do not speak for the first 90 seconds after issuing the prompt). After 3 minutes, cold-call 3 learners: 'Amina, what did you sketch and predict?' → 'Brian, do you agree with Amina — why or why not?' → 'Chebet, what is your prediction for which region is larger?' Record the three predictions verbatim on the board under the heading PREDICTIONS — we will test these. Do NOT confirm or deny any prediction at this stage.	By committing a prediction to paper before any instruction, learners activate prior knowledge of sectors and triangles, surface misconceptions (e.g., 'the curved bit must be bigger because it has the arc'), and create cognitive dissonance that motivates the evidence-gathering phase.	identify two regions (segment and triangle) — these learners are ready to lead peer discussion; (b) learners who confuse the segment with the sector — note their names for targeted questioning in Phase 2; (c) learners who leave the prediction sentence blank — prompt with 'What do you remember about how we split up the annular sector in Lesson 4?'
Observe Phase 📹 **VIDEO:** [Proof: radius is perpendicular to a chord it bisects] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Midpoints and Segment Bisectors (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each pair of learners receives: one paper circular disc (radius 7 cm, pre-drawn), a ruler, scissors, and a protractor. Step 1 — Mark a central angle of 80° and draw the sector (two radii + arc). Step 2 — Draw a chord connecting the two ends of the arc (the two points where the radii meet the circumference). Step 3 — Cut out the sector. Step 4 — Cut along the chord to physically separate the sector into (a) the isosceles triangle formed by the two radii and the chord, and (b) the curved segment (the 'bite' between the chord and the arc). Step 5 — Place each piece on squared centimetre paper and estimate its area by counting squares. Record: Triangle area (counted) = ___ cm²; Segment area (counted) = ___ cm². Step 6 — Using known formulas, calculate: Sector area = (80/360) × π × 7² = ___ cm²;	Before learners cut, display and read aloud: 'You are going to physically separate the sector into two pieces — a triangle and a curved segment — exactly the way the Laikipia farmer's fence separates the watered sector.' Demonstrate Step 1–2 on the board with a large paper disc. Ask: 'What formula did we use for sector area in Lesson 3?' (Cold-call 2 learners.) Ask: 'In Lesson 4 trigonometry, how did we find the area of a triangle when we know two sides and the included angle?' (Cold-call 2 learners; write ½ab sinC → ½r²sinθ on the board.) Then release pairs to work. During the activity, circulate and ask targeted questions: 'What do you notice about the sector area, the triangle area, and the segment area — how are the three numbers related?' (WAIT TIME: 12 seconds after asking, before accepting an	Concrete-Representational-Abstract (CRA) with physical decomposition. The cut-and-separate activity makes the abstract subtraction A_sector − A_triangle tangible and visible — learners can literally hold the segment in one hand and the triangle in the other. The grid-counting estimate then acts as an empirical check on the algebraic formula, reinforcing that mathematics describes physical reality.	Check three things as you circulate: (1) Is the chord drawn correctly (connecting the two arc endpoints, not a diameter)? (2) Is the triangle formula written as ½r²sinθ (using the central angle), not ½ × base × height? (3) Does the pair's formula-derived segment area fall within ±1 cm² of their grid count? Learners who have wrong chord placement need immediate re-teaching of the definition of a chord before proceeding.

	Triangle area by formula = ½ × 7^2 × sin80° = ___ cm^2. Step 7 — Compute: Segment area by subtraction = Sector area − Triangle area = ___ cm^2. Step 8 — Compare the formula-derived answer with the grid-counting estimate. Learners record all steps in their exercise books with a labelled diagram.	answer.) If a pair finishes early, prompt: 'What would happen to the segment area if the angle were 120° instead of 80° — would the segment grow or shrink relative to the triangle?' After 20 minutes, bring the class together. Cold-call one pair to present their numbers on the board. Ask the class: 'Does the subtraction match your grid count? What does that tell us?' Pause 10 seconds. Then write the general formula: $A_{segment} = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ and say: 'This is what the evidence just showed us — let's write it together.'		
Explain Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Midpoints and Segment Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four on two structured application problems, written on a task card. Problem 1 — Turkana Borehole Hatch: A water-project engineer in Turkana County has fabricated a circular iron hatch cover for a borehole. The cover has a radius of 35 cm. A straight locking bar (chord) is welded across the cover such that the central angle subtended by the bar is 110°. The engineer needs to weld a rubber seal only along the curved arc of the smaller region (the segment). (a) Calculate the area of the minor segment. (b) If rubber sealing foam costs Ksh 12 per cm^2, what is the cost of sealing the segment? Problem 2 — Sufuria Lid: A circular aluminium sufuria lid from Kamukunji Jua Kali market has a diameter of 28 cm. A straight crack runs across the lid, and the chord of the crack subtends a central angle of 72° at the centre. Calculate the area of the smaller region between the chord and the rim of the lid. After solving, each group writes a 'Formula Reasoning Statement': 'We subtracted the	Distribute task cards. Say: 'These two problems come straight from the real world — a Turkana borehole and a Kamukunji sufuria lid. Use the formula we derived from our paper-cutting evidence.' Allow 15 minutes for group work. Circulate and listen for reasoning. Prompt struggling groups: 'Which part of your sector is the triangle and which is the segment? Can you draw and label it first?' (WAIT TIME: after asking, step back for 10 seconds and let the group think before adding another prompt.) After 15 minutes, cold-call one group to present Problem 1 step-by-step on the board. Ask the class after the presentation: 'Does everyone agree with this group's use of $\frac{1}{2}r^2\sin\theta$ for the triangle? What would have gone wrong if they had used ½ × base × height instead?' (Cold-call 2 learners for responses.) Then cold-call a different group for Problem 2. After both presentations, facilitate a brief whole-class discussion: 'How does the segment formula connect back to the Laikipia farmer's challenge that started our unit? Where does	Contextualised problem solving with Formula Reasoning Statements. The Reasoning Statement ('We subtracted the triangle because…') forces learners to articulate the conceptual logic — not just execute the algorithm — making the decomposition strategy transferable. Connecting both problems back to the anchor phenomenon reinforces that the formula is a tool for real Kenyan contexts, not an abstract exercise.	Collect one written solution per group after presentations. Check for: (a) correct substitution of r and θ into both sub-formulas; (b) correct use of sin(110°) = sin(70°) for Problem 1 (since sinθ for obtuse angles — watch for learners who key sin(110°) directly into calculator vs. those who convert; both are correct but note the approach); (c) a coherent Reasoning Statement that mentions subtraction of the triangle. Groups whose Reasoning Statement says only 'because the formula says so' need follow-up probing in the DQB phase.

	triangle area because ______.' Groups then share their reasoning statements with the class.	the farmer need a segment calculation specifically?' (WAIT TIME: 12 seconds.)		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Midpoints and Segment Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB is displayed at the front. Each learner receives two sticky notes. On the first (green): write one thing the segment formula now answers about the Laikipia farmer's problem that the sector formula alone could NOT answer. On the second (orange): write one new question that today's lesson has raised — e.g., 'What if the chord creates a major segment — do we use a different formula?' Learners post their notes on the DQB in the columns: GREEN = 'We can now explain…' and ORANGE = 'We still wonder…'. The teacher reads 3–4 notes aloud, clusters related wonders, and previews: 'Some of you are asking about overlapping circles — that is exactly where we are going in Lesson 6.'	Give learners 4 minutes to write their sticky notes silently. Then invite them to post. Read aloud: (Green) 'Wangari writes: The segment formula tells us the area of the patch the farmer cannot water because the fence chord cuts across the sector — the sector formula would include the triangle the fence sits on.' Say: 'Excellent — Wangari has identified the exact practical value.' (Orange) 'Otieno writes: What about when the segment is more than half the circle — a major segment?' Say: 'Hold that thought — write it on the board as a starred wonder. We will revisit it in Lesson 6.' Cold-call one more learner from each column. Close the DQB update by pointing to the original driving question written at the top of the board: 'How can we calculate the exact area of any part of a circle?' and asking: 'Show of hands — who feels we have added a new tool to answer this question today?' Count and note for your records.	Driving Question Board curation. The DQB makes the cumulative arc of learning visible — each lesson's contribution is physically tracked. Green notes consolidate understanding; orange notes surface productive next-step questions, modelling authentic scientific inquiry practice (NGSS SEP 1: Asking Questions). Clustering the 'wonders' demonstrates that new knowledge generates new questions, normalising intellectual curiosity.	Read all green sticky notes after class. Check that learners are connecting the segment to the phenomenon (not just stating the formula). Learners whose green note says only 'We learned the segment formula' have not yet made the phenomenon connection — flag for individual follow-up at the start of Lesson 6. Count the number of orange 'wonders' that reference the major segment or overlapping circles — this signals readiness for Lesson 6 content.
Model Building Phase 📹 **VIDEO:** Proof: radius is perpendicular to a chord it bisects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Midpoints and	Learners retrieve their cumulative circle-decomposition diagram begun in Lesson 1 (a large annotated circle showing each sub-region unlocked lesson by lesson). Today they add: (a) a clearly labelled segment (shaded curved region between a chord and an arc) on their diagram; (b) the formula $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ written beside the segment with arrows showing the two components; (c) a brief annotation: 'Segment = Sector −	Say: 'Take out your circle model — the one we have been building since Lesson 1. You are going to add today's new region.' Display a completed teacher model on the board as a reference (do NOT show it until learners have attempted their own additions for 2 minutes). After 2 minutes, reveal the teacher model and say: 'Compare yours to this — what did you include that I didn't, or vice versa?' (WAIT TIME: 10 seconds.) Circulate during pair-share and	Cumulative annotated model revision (NGSS SEP 2: Developing and Using Models). The running diagram functions as a personal knowledge-building artefact — each lesson's addition is visually connected to previous sub-regions, reinforcing that sectors, annuli, annular sectors, and segments are all decompositions of the same circle. Anchoring the formula to a thumbnail sketch of the Turkana hatch makes the abstraction	Collect models at the end of the lesson (or photograph them). Check: (a) Is the segment correctly distinguished from the sector? (b) Are both sub-formulas (sector and triangle) shown as components? (c) Is the annotation 'Segment = Sector − Triangle' present? (d) Does the thumbnail sketch correctly shade the segment (not the whole sector)? Learners missing (b) need re-engagement with the derivation before Lesson 6.

Segment Bisectors (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Triangle'; (d) a small thumbnail sketch of the Turkana borehole hatch beside the formula, with the segment shaded, to anchor the formula to a real object. Learners then pair-share: 'Point to each region on your model and say its formula aloud to your partner — sector, triangle, segment.' The lesson closes with the teacher projecting the original Laikipia aerial photograph and asking: 'Point to where you would now use the segment formula on this farm map.'	listen for correct formula recitation. If a learner states the formula incorrectly, say: 'Say that again but tell me which part is the sector and which is the triangle' — do not immediately correct; scaffold the self-correction. Close with the aerial photograph prompt. Cold-call 2 learners to physically come to the board, point to the photograph, and name where the segment formula applies. Ask: 'If the farmer moves the fence chord further from the centre, what happens to the segment area — does it increase or decrease?' (WAIT TIME: 12 seconds; cold-call 1 learner for a final exit thought.)	concrete and memorable.	

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) During the paper-cutting activity, did learners correctly identify the chord and distinguish it from a diameter — if not, what additional vocabulary scaffolding is needed before Lesson 6? (2) In Problem 1, did the majority of learners handle sin(110°) correctly — if learners keyed sin(110°) into the calculator and got the same answer as sin(70°), did they understand WHY (supplementary angle identity), or were they algorithmically lucky? Plan a 2-minute clarification at the start of Lesson 6 if needed. (3) Were the Reasoning Statements ('We subtracted the triangle because…') conceptually grounded or merely procedural? If mostly procedural, increase the proportion of 'explain your reasoning' prompts in Lesson 6. (4) Did the DQB orange notes cluster around major segments and overlapping circles as anticipated — if learners are raising entirely different questions, revise the Lesson 6 entry point accordingly. (5) Did the cumulative model activity take longer than planned? If so, consider assigning the thumbnail annotation as a take-home task to protect in-class derivation time in future iterations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners physically cut a paper sector ($r = 7$ cm, $\theta = 80°$) along its chord, separated it into a triangle and a curved segment, estimated each area by counting squares on grid paper, then computed each area using formulas and compared the two methods.
What did I learn?	A segment is the region between a chord and its arc. Its area equals the sector area minus the triangle area: $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. This was confirmed when the formula-derived value matched the grid-count estimate to within measurement error.
How does this explain the phenomenon?	Learners applied the segment formula to find the rubber-seal area on a Turkana borehole hatch cover ($r = 35$ cm, $\theta = 110°$) and the curved crack region on a Kamukunji sufuria lid ($d = 28$ cm, $\theta = 72°$), connecting the formula back to the Laikipia farmer's challenge of calculating the exact area of a chord-cut region on the pivot field.

LESSON 6 (80 minutes): Area of a Common Region Between Two Intersecting Circles	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive and calculate the area of the common (lens-shaped) region formed when two circles intersect, and to apply this to overlapping irrigation plots and design contexts.
Knowledge	Learners will know that the common region between two intersecting circles (a lens or vesica shape) is composed of two circular segments, each calculated as the difference between a sector area and a triangle area, and that the total common area equals the sum of those two segments.
Skills	Learners will be able to: (i) identify the lens-shaped common region from two overlapping circles; (ii) use the distance between centres and the two radii to find the half-angle subtended at each centre using the cosine rule; (iii) calculate each segment area as $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$; (iv) sum the two segment areas to obtain the total common area; (v) apply the formula to real-life overlapping irrigation and design problems.
Attitudes	Learners will appreciate the value of precise mathematical decomposition in avoiding waste and ensuring fairness — e.g. so that two Laikipia farmers whose pivot systems overlap can agree on exactly how much land is jointly watered and plan their water budgets without conflict.
Key Inquiry Question	How do we determine the area of the common region between two intersecting circles, and how is this applied in real-life situations such as overlapping irrigation plots and school art stencils?
Purpose in Storyline	In Lessons 1–5, learners found the areas of annuli, sectors, annular sectors, and single circular segments — giving the Laikipia farmer tools for the ring between two pipe radii and for wedge-shaped swept zones. Today's lesson tackles the final puzzle: when two adjacent centre-pivot irrigation circles overlap, what is the exact area of the shared lens-shaped patch? Resolving this lets both farmers avoid double-counting watered land and correctly split their water-use costs. It also unlocks the school art-project stencil (two overlapping Kericho tea-tray circles), completing the learners' toolkit for decomposing any circle-based region.
Safety Notes	Compasses have sharp points — learners must keep compass tips pointing downward when not in use and must not wave compasses in the air. Scissors used for circular cut-outs must be handled with blades closed when passing to a neighbour. Remind learners to keep desks clear of bags to avoid tripping when moving between groups.

B. LESSON OVERVIEW

Learners open by predicting the area of the lens-shaped overlap between two circular irrigation plots drawn on a grid. They then use pre-cut circular card stencils (the 'Kericho tea-tray stencils') to observe and physically trace the common region, measure the relevant distances, and record data. In the Explain phase, the teacher guides structured sensemaking: learners decompose the lens into two segments, apply the cosine rule to find the central angles, then compute each segment area using the formula $A_{segment} = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$. They add the two segment areas to find the total lens area. The DQB is updated to flag that all major sub-regions of a circle have now been covered. Learners close by building a revised cumulative model of the farmer's field showing the labelled lens region alongside the annulus, sector, and segment from prior lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a projected aerial-style diagram of two overlapping circular irrigation plots in Laikipia: Circle A (centre pivot at Borehole A, radius 42 m) and Circle B (centre pivot at Borehole B, radius 35 m), with the distance between the two boreholes being 49 m. The overlapping lens region is shaded green and labelled 'shared watered zone'. Working individually for 4 minutes in their exercise books, learners write: (a) a rough sketch of the lens and label what they know; (b) a prediction in square metres of the shared area, with a one-sentence justification of their reasoning strategy. Learners then share predictions with a partner (Think-Pair) for 2 minutes before three predictions are cold-called to the class.	Display the aerial diagram and say: 'Two farmers near Rumuruti in Laikipia have set up adjacent centre-pivot systems. Borehole A waters a circle of radius 42 metres. Borehole B waters a circle of radius 35 metres. The boreholes are 49 metres apart. The green lens is the land both systems water at the same time. In your books, sketch this situation and predict: how many square metres is the shared green zone? Write your reasoning — do not just write a number.' Allow 4 minutes of silent individual work. WAIT TIME: do not prompt during the 4 minutes. After pair-share, cold-call exactly 3 learners: 'Achieng, what did you and your partner predict and why?', 'Kipkoech, yours?', 'Zawadi, do you agree or disagree with what you heard?' Record the three predictions on the board in a 'Prediction Bank' column. Say: 'We will return to these predictions at the end of today's lesson to see who was closest and why.'	Predict-before-instruction (activation of prior knowledge): By committing a prediction to paper before any instruction, learners surface their current mental models — whether they attempt to use full-circle subtraction, rectangle approximation, or recall of segment work from Lesson 5. Surfacing misconceptions (e.g. 'just subtract the two circle areas') gives the teacher real-time diagnostic information and creates cognitive dissonance that motivates the Observe phase.	Circulate during the 4-minute prediction window. Observable indicators of readiness: (i) learner sketches show two overlapping circles with a shaded lens — not just one circle; (ii) learner references a strategy from prior lessons (sector, segment) rather than guessing randomly. Flag learners who write '$\pi r_1^2 - \pi r_2^2$' (subtracting full circle areas) — these learners hold a key misconception to address explicitly in the Explain phase. Collect three exercise books from flagged learners for follow-up.
Observe Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar	Groups of four receive: two circular card cut-outs per pair (pre-cut 'Kericho tea-tray stencils' — Circle A: radius 7 cm, Circle B: radius 5.5 cm, representing the 42 m and 35 m plots at 1 cm : 6 m scale), a ruler, a protractor, a pencil, and a plain A3 sheet. Activity steps (15 minutes): (1) Place Circle A on the A3 sheet and trace it. Mark centre A. (2) Measure 7 cm (scaled 42 m borehole-to-borehole distance becomes 49/6 ≈ 8.2 cm — teacher directs learners to use 8.2 cm) from centre A and mark centre B. (3) Overlap Circle B so its centre is at B and trace Circle B. Shade the	Distribute materials before the lesson starts so no time is lost. As groups work, circulate and ask probing questions — do NOT give answers: 'What is the name of the region between chord PQ and the arc of Circle A?' (expected: segment). 'How did we find segment area in Lesson 5?' After 8 minutes, pause the class with a hand signal and say: 'Everyone freeze. Look at your lens. Can someone tell me — without looking at notes — what two shapes make up that green region?' WAIT TIME: 12 seconds. Cold-call 2 learners. Then say: 'Correct — two	Concrete manipulatives with guided recording (CRA — Concrete stage): Physical overlapping of card cut-outs makes the abstract lens region tangible. The structured results table scaffolds precision and creates a shared data set that all groups can compare. The three guided questions at the back are a 'bridge' strategy — they explicitly connect the new shape (lens) to the known shape (segment from Lesson 5), activating prior schema before formal instruction begins.	Check results tables for two key indicators: (i) learners correctly identify AP = AQ = 7 cm (radii of Circle A) and BP = BQ = 5.5 cm (radii of Circle B), confirming understanding that radii are equal; (ii) guided question 2 responses mention 'segment' — if fewer than half the groups write 'segment', pause instruction and briefly re-display the Lesson 5 segment diagram before proceeding to Explain. Listen for groups that describe the lens as 'two triangles' — redirect by asking them to look at the curved boundary.

readings	lens region. (4) Mark the two intersection points P and Q. Draw chord PQ. (5) Use a ruler to measure AP, AQ, BP, BQ and the chord length PQ. Record all measurements in a results table with columns: [Measurement | Scaled value (cm) | Real-world value (m)]. (6) At the back of the sheet, learners write answers to three guided questions: 'What two shapes make up the lens?', 'Which previous lesson shape does each piece remind you of?', 'What extra information would you need to calculate the area of each piece?'	segments, one from each circle. Now finish your measurements and answer the three questions on the back.' In the final 2 minutes of Observe, ask one group to hold up their A3 sheet and explain their sketch to the class in 30 seconds.		
Explain Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class structured instruction with worked examples and peer practice (25 minutes). Part A — Deriving the formula (10 min): The teacher works through the Laikipia example step-by-step on the board, narrating each step aloud. Learners copy the derivation into their books. Step 1: Label — Circle A (radius r_1 = 42 m, centre A), Circle B (radius r_2 = 35 m, centre B), distance AB = d = 49 m, intersection points P and Q. Step 2: Find angle PAQ (= 2α) using the cosine rule in triangle APB: $\cos\alpha = (r_1^2 + d^2 - r_2^2) / (2 \cdot r_1 \cdot d) = (42^2 + 49^2 - 35^2) / (2 \cdot 42 \cdot 49) = (1764 + 2401 - 1225) / 4116 = 2940/4116 \approx 0.7143$; $\alpha \approx 44.4°$; so angle PAQ = $2\alpha \approx 88.8°$. Step 3: Find angle PBQ (= 2β): $\cos\beta = (r_2^2 + d^2 - r_1^2) / (2 \cdot r_2 \cdot d) = (1225 + 2401 - 1764) / 3430 = 1862/3430 \approx 0.5430$; $\beta \approx 57.1°$; so angle PBQ = $2\beta \approx 114.2°$. Step 4: Area of segment from Circle A = $(2\alpha/360)\cdot\pi\cdot r_1^2 - ½\cdot r_1^2\cdot\sin(2\alpha) = (88.8/360)\cdot\pi\cdot 42^2 - ½\cdot 42^2\cdot\sin(88.8°) \approx 0.2467\cdot 5541.8 - 0.5\cdot 1764\cdot 0.9997 \approx 1367.7 - 881.9 \approx 485.8$ m². Step 5: Area of	During Part A derivation, narrate explicitly: 'I am using the cosine rule — which you met in Trigonometry — because I have three sides of triangle APB and need an angle. Watch how the angle at A tells me the sector slice of Circle A that contains my segment.' After Step 3, pause and say: 'Before I go further — what is the formula for the area of a segment? Do not look at your notes.' WAIT TIME: 15 seconds. Cold-call 3 learners. Write the formula on the board only after the cold-calls. After Step 6, say: 'The farmer can now tell her neighbour exactly how much land — 1 148 square metres — both their systems water simultaneously. She uses this to share the water bill fairly.' During Part B pair work, circulate and use targeted prompts only: 'Show me your cosine rule substitution for angle C.' 'Is angle PBQ the angle you want for the segment formula, or is it the half-angle?' Do NOT give answers. When the presenting pair writes on the board, ask the class: 'Pause — before they go to Step 4, thumbs:	Worked example with narrated reasoning (I Do — We Do — You Do): The teacher's narrated derivation makes the cognitive process visible ('I am doing X because Y'). Connecting to the cosine rule from Sub-Strand 2.4 (Trigonometry) is a cross-strand transfer strategy — it shows learners that mathematical tools recur across topics. The Prediction Bank revisit creates a closing loop of cognitive dissonance resolution. The 'Thumb Criteria' during peer presentation is a low-stakes whole-class formative check that keeps all learners accountable during the presentation, not just the presenters.	During Part A: if fewer than 2 of 3 cold-called learners can state the segment formula correctly, re-teach the formula from Lesson 5 before proceeding (have the Lesson 5 formula card ready as a contingency). During Part B: look for two error types — (i) learners using the half-angle (α or β) instead of the full angle (2α or 2β) in the segment formula — prompt: 'Which angle is at the centre of the sector?'; (ii) learners adding two sector areas without subtracting the triangles — prompt: 'What is the difference between a sector and a segment?' Collect two pairs' written solutions for marking before the next lesson.

	segment from Circle B = $(2\beta/360)\cdot\pi\cdot r_2^2 - ½\cdot r_2^2\cdot\sin(2\beta)$ = $(114.2/360)\cdot\pi\cdot 35^2 - ½\cdot 35^2\cdot\sin(114.2°) \approx 0.3172\cdot 3848.5 - 0.5\cdot 1225\cdot 0.9114 \approx 1220.5 - 558.2 \approx 662.3$ m². Step 6: Total lens area = 485.8 + 662.3 ≈ 1148.1 m². Teacher returns to Prediction Bank: 'Which prediction was closest? Let us honour that mathematician.' Part B — Peer practice (15 min): Learners work in pairs on a new problem: Two overlapping circular tea-estate plots near Kericho — Plot C radius 28 m, Plot D radius 21 m, centres 35 m apart. Find the area of the common region. Pairs must show all six steps. After 12 minutes, one pair presents their solution on the board. The class uses a 'Thumb Criteria' check (thumb up / sideways / down) at each step.	do you agree with their angle PBQ?' WAIT TIME: 10 seconds, then survey thumbs before allowing the pair to continue.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intercepts from an equation](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front. Each learner receives one sticky note. They write one of the following (their choice): (a) a claim — something they can now answer from the driving question that they could not before; (b) a connection — how today's lens formula links to a formula from a previous lesson; or (c) a remaining question — something about circle-area problems they still cannot yet solve. Learners stick their notes in the appropriate column (CLAIM / CONNECTION / STILL WONDERING). The teacher reads three notes aloud and the class votes with a show of hands whether each claim is fully supported by today's evidence.	Say: 'We are now six lessons into our investigation for the Laikipia farmer. Look at the driving question: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste? Take your sticky note. You have 3 minutes — write a CLAIM, a CONNECTION, or a STILL WONDERING. Be specific: name the shape and the formula.' WAIT TIME during writing: 3 minutes uninterrupted. As you read three notes aloud, ask: 'Class — hands up if you agree this claim is fully proven by today's work.' For any 'STILL WONDERING' notes about composite regions or regions involving three circles, say: 'Excellent — that is a mathematician's instinct. Flag that for our model-building session.'	Three-column DQB sort (Claim / Connection / Still Wondering): This strategy makes the progression of understanding visible across all eight lessons. Sorting into three columns — rather than a flat list — forces learners to distinguish between settled knowledge (claim), structural understanding (connection), and productive uncertainty (still wondering). The public vote on claims introduces peer accountability and scientific discourse norms.	Observable indicators: (i) CLAIM notes should reference the lens/common-region formula specifically — vague claims like 'I understand circles now' are redirected: 'Can you name the exact formula?'; (ii) CONNECTION notes that link to segment (Lesson 5) or sector (Lesson 2/3) show strong schema integration — celebrate these publicly; (iii) STILL WONDERING notes about three overlapping circles or irregular overlaps signal high-readiness learners who can be given an extension task in Lesson 7.

Model Building Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative field model — a large annotated diagram of the Laikipia farmer's circular plot that has been updated in each prior lesson. Today they add: (1) a second overlapping circle representing a neighbouring farmer's pivot system, with the lens-shaped common region shaded in a new colour (green) and labelled with the formula: $A\_lens = [(2\alpha/360)\pi r_1^2 - ½r_1^2\sin 2\alpha] + [(2\beta/360)\pi r_2^2 - ½r_2^2\sin 2\beta]$; (2) a legend entry for the lens alongside the existing legend entries for annulus (Lesson 1), sector (Lessons 2–3), annular sector (Lesson 4), and segment (Lesson 5); (3) a written note in the margin: 'The farmer can now calculate the exact watered area of every sub-region of her field. She can plan planting rows, fencing, and water costs accurately.' Learners write a two-sentence 'Model Upgrade Note' explaining what new information this lesson's formula adds to the farmer's planning toolkit that she did not have after Lesson 5.	Distribute the cumulative model sheets (or direct learners to open their model portfolios). Say: 'Your model has grown across six lessons. Today you are adding the final major sub-region: the lens. Add it to your diagram now — use green shading, label the formula, and update the legend. You have 8 minutes.' Circulate and check that: (i) learners are drawing a second circle that genuinely overlaps (not just touching); (ii) the formula label includes both segment terms; (iii) the Model Upgrade Note is specific — redirect vague notes: 'Tell me something the farmer can do NOW that she could not do after Lesson 5.' In the final 2 minutes, cold-call 2 learners to read their Model Upgrade Note aloud. Say: 'Next lesson we will look at how to handle composite problems — combining two or more of the sub-regions we have learned. Your model will need to be ready.'	Cumulative annotated model revision: The running field diagram serves as a 'knowledge artefact' — a single visual that accumulates and integrates all sub-region formulae across the unit. Each lesson's addition makes the progression of understanding concrete and reviewable. The Model Upgrade Note forces metacognitive articulation: learners must state not just what they drew but why it advances the farmer's problem. This mirrors the scientific practice of revising models in light of new evidence (NGSS SEP 2: Developing and Using Models).	Collect 5 cumulative model sheets (selected randomly) for teacher review before Lesson 7. Look for: (i) all five sub-regions correctly labelled with their formulae (annulus, sector, annular sector, segment, lens); (ii) the lens formula includes both segment areas — a single-segment formula is a misconception to address at the start of Lesson 7; (iii) Model Upgrade Notes that articulate a specific planning benefit (e.g. 'splitting water costs with a neighbour') rather than a restatement of the formula. Use these five sheets to group learners for the Lesson 7 composite-problem task.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and note responses in your professional journal: (1) Prediction Bank: How spread out were the three cold-called predictions? Did any learner correctly identify the two-segment structure before instruction — and if so, how might you use that learner as a peer explainer in a future lesson? (2) Cosine rule recall: Were learners able to apply the cosine rule fluently, or did the cross-strand transfer cause significant hesitation? If more than one-third of learners struggled at Step 2, plan a 5-minute cosine-rule recap at the start of Lesson 7 before the composite task. (3) Common errors: Did learners confuse the half-angle (α) with the full central angle (2α) when substituting into the segment formula? Document how many pairs made this error during Part B and whether your in-flight prompt ('Which angle is at the centre of the sector?') was sufficient. (4) DQB quality: Were CLAIM notes specific enough to reference the lens formula by name? If not, consider adding a sentence stem to the sticky-note prompt in future: 'I can now calculate ___ using the formula ___.' (5) Equity check: Did all learners actively participate in the card-manipulation activity, or did one member of each group dominate the tracing task? Adjust group roles for Lesson 7 (assign a 'materials manager', 'recorder', 'presenter', and 'checker' explicitly). (6) Phenomenon connection: Did learners spontaneously mention the Laikipia farmer during the Explain phase, or did the context feel detached from the mathematics? If detached, open Lesson 7 by re-reading the farmer's water-budget problem aloud before launching the composite task.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners overlapped two circular card cut-outs (Kericho tea-tray stencils) on an A3 sheet, traced the lens-shaped common region, identified the two intersection points P and Q, drew chord PQ, and measured the radii and centre-to-centre distance. They recorded scaled and real-world values in a results table and identified the lens as being composed of two circular segments.
What did I learn?	The common region between two intersecting circles is a lens shape equal to the sum of two circular segments — one from each circle. The central angle for each segment is found using the cosine rule applied to the triangle formed by the two centres and one intersection point: $\cos\alpha = (r_1^2 + d^2 - r_2^2)/(2r_1d)$ and $\cos\beta = (r_2^2 + d^2 - r_1^2)/(2r_2d)$. Each segment area is calculated as $A = (2\theta/360)\pi r^2 - \frac{1}{2}r^2\sin(2\theta)$, and the total lens area is the sum of both segment areas.
How does this explain the phenomenon?	For the Laikipia farmer's two overlapping irrigation plots ($r_1 = 42$ m, $r_2 = 35$ m, $d = 49$ m), the central angle at A is approximately 88.8° and at B approximately 114.2°. The segment from Circle A has area ≈ 485.8 m², the segment from Circle B ≈ 662.3 m², giving a total shared (lens) area of approximately 1 148 m². This tells both farmers exactly how much land is jointly watered, enabling a fair split of water costs and preventing double-counting in planting and fencing plans.

LESSON 7 (80 minutes): Mixed Application and Problem Solving: Combining Circle-Part Formulae	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners integrate all circle-part formulae — annulus, sector, segment, annular sector, and common region — to solve multi-step real-life problems drawn from Kenyan agricultural, engineering, and artisan contexts, thereby advancing the farmer's planning challenge at the heart of the unit phenomenon.
Knowledge	Learners know and can accurately state the formulae for the area of an annulus ($\pi(R^2-r^2)$), a sector ($\theta/360 \times \pi r^2$), a segment (sector area − triangle area), an annular sector ($\theta/360 \times \pi(R^2-r^2)$), and the common region between two intersecting circles; and can identify which formula or combination of formulae applies to a given composite shape.
Skills	Learners can (a) decompose a composite circle-part figure into identifiable sub-regions; (b) select and apply the correct formula(e) for each sub-region; (c) combine sub-region areas correctly using addition and subtraction; (d) present a structured multi-step solution; and (e) peer-assess a classmate's solution against a clear criterion.
Attitudes	Learners appreciate that mathematical precision directly prevents resource waste and supports fair land-use planning, and they value collaborative problem solving and honest peer feedback as tools for improving accuracy.
Key Inquiry Question	How do we determine and apply the areas of parts of a circle to solve complex, multi-step real-life problems?
Purpose in Storyline	In previous lessons the farmer identified individual sub-regions of the Laikipia pivot-irrigation field — the full circle, the annulus between two pipe rings, the swept sector, and the segment of overlap between adjacent plots. Today, the farmer must make complete, integrated plans: How much fencing is needed around an annular sector plot in the Rift Valley? How many maize seedlings fit in a Nairobi urban farm shaped like a circle minus a segment? How does a Mombasa fish-pond designer calculate the shaded region between two intersecting circular pools? Learners now wield every tool they have built and combine them — exactly as a real engineer or land planner would — to answer the driving question in full.
Safety Notes	No hazardous materials are used. Compasses have sharp points — learners must cap them when not in use and must not gesture with open compasses. Calculator screens should not be shared in ways that tempt copying; each learner must record their own working.

B. LESSON OVERVIEW
Lesson 7 of 8 in the evidence-gathering sequence. Learners work in groups of four to solve four multi-step, real-life problems that each require combining two or more circle-part formulae. Problems are set in Nairobi, Mombasa, Rift Valley, and Laikipia contexts, and each connects explicitly to the anchoring phenomenon. The lesson follows the ARES five-phase structure: Predict (individual written prediction on a composite diagram), Observe (structured group problem-solving with worked-example scaffold), Explain (group solution presentations with peer assessment using a structured rubric), Driving Question Board update (adding new insight cards), and Model Building (learners revise their cumulative circle-decomposition model to incorporate composite shapes). Peer assessment using the KICD four-level rubric language (Exceeds / Meets / Approaches / Below Expectations) is embedded in the Explain phase.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner receives a single A5 card showing a composite diagram: a large circle of radius R = 14 m with a smaller concentric circle of radius r = 7 m, and a sector of angle 120° shaded within the annular ring, plus a triangular segment cut away from the inner circle. Beneath the diagram the prompt reads: 'This is a section of the Laikipia farmer's field. Without calculating, write (a) which formulae you think are needed, (b) the order in which you would use them, and (c) your rough estimate of the shaded area if the full large circle has area ≈ 616 m².' Learners write individually for 5 minutes, then share predictions with their neighbour for 2 minutes. Teacher takes a 3-person cold-call show of hands to hear three different predicted approaches before the group work begins.	Distribute the predict card face-down. Say: 'Do NOT turn it over until I say go.' After 30 seconds of silent reading time, say: 'You have 5 minutes — write your own thinking. No calculators yet.' Set a visible timer. Circulate and note 2–3 interesting or divergent predictions without correcting them. After individual time, say: 'Turn and share with your neighbour for 2 minutes — listen to hear if they chose a different formula order.' After sharing, cold-call exactly 3 learners using a name-stick draw. For each, ask: 'What formula did you put FIRST and why?' Apply 12-second wait time before accepting the answer. Record the three different approaches on the board as 'Prediction A', 'Prediction B', 'Prediction C' and say: 'We will return to these at the end of the lesson to see who was closest.'	Activation of Prior Knowledge via Predict-Before-Instruction: By asking learners to sequence formulae before any new instruction, the teacher surfaces existing mental models of when each formula applies. Recording divergent predictions publicly creates a reference point that motivates careful evidence gathering and gives learners a concrete claim to confirm or revise — directly mirroring the farmer's need to plan before purchasing materials.	Observable indicator: Each learner's predict card contains at least two named formulae (e.g., 'annular sector', 'segment') and a written rationale for their order. Teacher scans cards during circulation — if fewer than 60% of learners name two or more formulae, pause and ask: 'Look at the diagram — how many separate shaded regions do you count?' to re-anchor decomposition thinking before moving on.
Observe Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Groups of four receive a Problem Set card containing four problems, one per Kenyan context. Each problem requires combining at least two formulae. Groups are assigned one problem as their 'lead problem' but must attempt all four. A 'Formula Reference Strip' (listing all five formulae with diagrams, no worked examples) is available face-up on each desk. Calculators and geometrical sets are available. PROBLEM 1 — RIFT VALLEY MAIZE PLOT (Annular Sector + Segment): A centre-pivot irrigation arm in Nakuru County sweeps through an angle of 150°. The outer pipe radius is R = 28 m and the inner pipe radius is r = 14 m. A	Before releasing the problem cards, say: 'Your job is to act like the engineer advising the Laikipia farmer — you must show every step so the farmer can check your working.' Distribute materials simultaneously to all groups. For the first 3 minutes, do NOT intervene — allow productive struggle. Then begin a circulation loop, visiting each group for 90 seconds. Use these targeted questions: — If a group is stuck on decomposition: 'How many separate regions do you see? Draw a boundary around each one and name it.' — If a group has set up the correct formula but wrong substitution: 'Read your formula aloud — what	Structured Group Problem Solving with Decomposition Scaffolding: The Formula Reference Strip ensures learners focus cognitive effort on selecting and sequencing formulae (higher-order thinking) rather than recalling syntax. The four problems are deliberately ordered by increasing structural complexity (single composite → two-region → common region → multi-part fencing plan), creating a gradient that lets all groups experience early success before tackling harder problems. This mirrors the engineering practice of breaking a complex system into manageable sub-problems — exactly the skill the farmer needs.	Observable indicators: (1) Each group's manila paper shows a labelled diagram with sub-regions named and a step-by-step formula chain before a numerical answer. (2) During teacher circulation, at least 3 of 4 groups can verbally explain which formula they used and why, without prompting from the reference strip. If a group cannot name the sub-regions, teacher draws a blank decomposition diagram beside their work and asks: 'Fill in the name of each region — then decide which formula matches each name.'

	triangular bund (earth wall) cuts a segment from the inner circle; the chord of the segment is 14 m and the perpendicular height from the chord to the arc is 3.5 m. Find the exact area of maize that receives water (annular sector minus the segment). [Use π = 22/7.] PROBLEM 2 — NAIROBI URBAN FARM (Full Circle minus Segment): A circular urban farm plot in Kibera, Nairobi, has radius 21 m. A straight road cuts across it; the chord of the road is 21 m. Find the area of the larger region of the farm that remains usable for growing sukuma wiki. [Use π = 22/7; establish triangle type from chord = radius.] PROBLEM 3 — MOMBASA FISH PONDS (Common Region of Two Intersecting Circles): Two circular fish ponds near Mtwapa, Mombasa, each have radius 10 m. Their centres are 10 m apart. Find the area of the region where the ponds overlap (the common region), giving the answer in terms of π and in surd form where needed. [Hint: Each contributing sector has angle 120°; use the two-sector-minus-two-triangle method.] PROBLEM 4 — LAIKIPIA COMPOSITE FENCING PLAN (Annular Sector + Full Sector): The farmer in Laikipia wants to fence a composite plot made of (i) an annular sector of outer radius 42 m, inner radius 21 m, and angle 90°, joined to (ii) a full sector of radius 21 m and angle 90° pointing inward. Find the total area	does R represent in the diagram in front of you?' — If a group finishes early: 'Have you verified your answer makes sense? Could the shaded area ever be LARGER than the full outer circle? Check yours.' At the 18-minute mark, give a 2-minute warning: 'Two minutes — if you haven't reached a final answer, write down the furthest step you reached and explain in words what you would do next.' Cold-call one member from each group (using name sticks) to read their solution summary aloud before the Explain phase begins. Apply 10-second wait time after each cold-call.		

	enclosed and determine how many bags of fertiliser are needed if one bag covers 120 m². [Use π = 22/7.] Groups work through their lead problem fully, showing all steps. They then attempt the remaining problems. After 20 minutes of group work, each group is given 3 minutes to write a 'solution summary' (answer + key formula chain) on a large sheet of manila paper for display.			
Explain Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups post their manila-paper solution summaries on the classroom walls (gallery style). Each group appoints a 'presenter' and a 'scribe'. Groups rotate around the gallery in 3-minute intervals — each group visits every other group's solution. During each visit, the visiting group's scribe completes a Peer Assessment Card using KICD rubric language: they award a level (Exceeds / Meets / Approaches / Below Expectations) for (a) correct formula selection, (b) correct substitution and working, and (c) correct final answer and units, and they must write one specific improvement suggestion using the stem: 'To improve this solution, you could…'. After the gallery rotation (12 minutes), each group returns to their own solution and has 3 minutes to read all peer feedback and decide: 'Do we stand by our answer, or do we want to revise it — and why?' A spokesperson from each group gives a 30-second verbal verdict to the class. Teacher facilitates a whole-class 5-minute convergence on correct answers for all four problems, surfacing the most	Before the gallery rotation, model the peer assessment task using a deliberately flawed example on the board (e.g., a solution that used the full-circle formula instead of the annular sector formula). Say: 'I am going to assess this solution. Watch how I use the rubric language.' Model writing: 'Formula selection — Below Expectations, because the full-circle formula was used instead of the annular sector formula. To improve, you could draw the sub-regions first and label each one before choosing a formula.' Then say: 'Now you will do exactly this for your peers — be specific and honest, because in engineering, a vague comment cannot fix a design.' During the gallery rotation, circulate and listen for misconceptions. Note the most common error type (e.g., forgetting to subtract the triangle in a segment calculation). At the convergence stage, cold-call 4 learners (one per problem) to give the correct answer and formula chain. Apply 12-second wait time. If a correct answer is given, ask a second cold-call: 'Can you tell us	Gallery Walk with Structured Peer Assessment: The physical rotation of groups past each other's work externalises mathematical reasoning and creates a low-stakes accountability structure. Using KICD's own four-level rubric language for peer feedback aligns the language learners use with the language of formal assessment, building metacognitive awareness. The 'stand by or revise' moment is a deliberate claim-evidence-reasoning (CER) sensemaking move: learners must publicly commit to a position and justify it with evidence from the peer feedback — the same epistemic practice a farmer or engineer uses when reviewing a contractor's plan.	Observable indicators: (1) Every Peer Assessment Card contains a specific rubric level with a written justification (not just a tick). (2) During the 30-second verdict, the spokesperson references at least one piece of peer feedback by name ('Our peers noted that…'). (3) During whole-class convergence, at least 80% of learners can identify the correct formula chain for their group's lead problem when cold-called. If a group's verdict contains no reference to peer feedback, teacher prompts: 'Show me one card — what did they write for formula selection? Does that match what you did?'

	common errors.	the ONE step that is most often done wrong in this problem?' Conclude by returning to the three 'Prediction' labels on the board from Phase 1 and asking: 'Whose prediction was closest? What did they get right that the others missed?' Take a show of hands.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (different colours). On the first (green): they write one new thing they can now confidently do that they could not do at the start of the lesson (e.g., 'I can decompose a composite shape into named sub-regions and apply two formulae in sequence'). On the second (orange): they write one remaining question or uncertainty about applying circle-part formulae in real life (e.g., 'I still wonder — how do engineers handle regions where three circles overlap?'). Learners post both notes on the class Driving Question Board under the column 'Lesson 7 — Multi-Step Problems'. The teacher reads three orange notes aloud and asks: 'Will our final lesson be able to answer any of these?' Learners indicate with a thumbs-up/thumbs-down.	Distribute sticky notes and say: 'You have 3 minutes — individual thinking, no sharing yet.' After writing, direct learners to the DQB and say: 'Green notes go under WHAT I CAN NOW DO. Orange notes go under WHAT I STILL WONDER.' Once posted, read three orange notes selected from different corners of the board (to ensure range). For each, ask the class: 'On a scale of one to five fingers, how many of you share this question?' Count fingers. Say: 'These are exactly the questions a real engineer would ask before signing off on a design. In Lesson 8 we will tackle the hardest of these in a formal assessment context.' Point explicitly to the DQB column for Lesson 8 and say: 'Your orange notes are helping us design that final lesson together.'	Two-Colour DQB Protocol (Can Do / Still Wonder): Separating 'confidence' notes from 'uncertainty' notes gives learners a structured metacognitive routine that distinguishes consolidated knowledge from open questions. This directly mirrors scientific practice — a researcher distinguishes findings from future research questions. For the phenomenon, it positions learners as genuine co-investigators: their orange-note questions about three-circle overlaps or irregular-sector boundaries are authentic extensions of the farmer's planning challenge, reinforcing that mathematics is an open, growing tool rather than a closed list of procedures.	Observable indicators: (1) Every learner posts at least one green and one orange note (100% participation check — teacher counts posted notes vs. learner count). (2) Green notes reference a specific formula or skill (not vague statements like 'I learned a lot'). (3) Orange notes are framed as genuine mathematical or real-life questions, not complaints. If a learner's green note is vague, teacher prompts quietly: 'Name the formula you used today that you couldn't use last week.'
Model Building Phase 📹 **VIDEO:** [Intercepts from an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal]	Learners retrieve their personal 'Circle Decomposition Model' — a cumulative annotated diagram they have been building across all seven lessons. Today's task: on a new section of the model labelled 'Lesson 7 — Composite Shapes', each learner draws a composite figure of their own invention (inspired by any of the four problems) that combines at least two circle-part sub-regions. They label each sub-region with its name and formula, write the	Say: 'Your model is your growing map of the farmer's problem. Today you are the designer — invent a composite shape that the farmer might actually encounter.' Give 10 minutes for individual drawing and annotation. Circulate with a focus on formula labelling: if a learner has written a formula without a diagram label, ask: 'Point to the part of your diagram where R stops and r starts.' Do not correct errors verbally — instead ask a question that makes the	Generative Modelling with Learner-Invented Composites: Asking learners to invent their own composite figure rather than reproduce a teacher-given one is a high-leverage sensemaking move — it requires genuine understanding of the structural logic of decomposition, not just procedural recall. The sentence connecting the invented shape to the farmer phenomenon ensures the model remains anchored to real-world meaning rather than	Observable indicators: (1) Every learner's Lesson 7 model section contains a labelled composite diagram with at least two named sub-regions and their corresponding formulae. (2) The combined-area expression is algebraically correct (e.g., Area = $\theta/360 \times \pi(R^2-r^2) - \frac{1}{2}r^2\sin\theta$). (3) The connecting sentence names a specific farm feature (pipe ring, sector, bund, fish pond, etc.). Exit-ticket check: as learners pack up, teacher asks 3 cold-called learners

[Multiplication (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	combined-area expression (not necessarily a numerical answer), and add one sentence connecting their composite figure to the Laikipia farmer's scenario: 'This shape could represent [part of the farm] because…'. Learners who finish early add a second composite figure using a different formula combination. In the final 3 minutes, learners pair-share their invented composite figure with a partner and check each other's formula labels for accuracy.	error visible. At the 10-minute mark, say: 'Stop drawing — pair up with the person next to you. You have 2 minutes each to explain your shape and check your partner's formula labels. If you spot an error, circle it in pencil and write a question mark — do not erase it, let your partner fix it.' In the final 2 minutes, cold-call 2 learners to share their invented shape with the class. For each, ask: 'How does this shape help the Laikipia farmer?' Apply 10-second wait time. Close with: 'In our final lesson — Lesson 8 — you will use everything in your model to tackle a formal multi-step assessment problem. Make sure your model is complete and corrected tonight.'	becoming an abstract exercise. The pair-check step embeds peer error-detection — a core engineering-design practice — and gives learners agency over the accuracy of their own tools heading into the final lesson.	to say their combined-area expression aloud. If an expression is incorrect, teacher says: 'Your partner circled something — look at it now and tell me what you would change', giving 15 seconds of wait time before offering a targeted hint.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes in your professional journal:

1. DECOMPOSITION FLUENCY: During Phase 2, could most groups correctly identify and name all sub-regions of their problem without teacher prompting? If not, which region type (annular sector, segment, common region) caused the most confusion, and how will you address this in Lesson 8's formal assessment scaffold?

2. PEER ASSESSMENT QUALITY: Did learners' Peer Assessment Cards contain specific, formula-level feedback, or were they vague ('good job', 'wrong answer')? If feedback was vague, consider introducing a sentence-stem card in Lesson 8: 'The formula used was ___, but it should have been ___, because ___.'

3. FORMULA COMBINATION ERRORS: During the whole-class convergence, what was the single most common multi-step error (e.g., adding instead of subtracting a segment, using diameter instead of radius after combining formulae)? Prepare one targeted warm-up question for Lesson 8 that directly addresses this error.

4. PHENOMENON CONNECTION: When learners presented their 'stand by or revise' verdicts, did they voluntarily reference the Laikipia farmer scenario, or did the connection feel forced? If voluntary references were rare, consider opening Lesson 8 with a brief re-viewing of the aerial photograph and asking: 'Which of today's four problems maps most directly onto what the farmer actually needs to know?'

5. MODEL COMPLETENESS: How many learners' Circle Decomposition Models now contain entries for all seven lessons? Identify learners with gaps and plan a 5-minute peer model-review at the start of Lesson 8 to fill missing sections before the formal assessment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined four composite circle-part diagrams set in Rift Valley, Nairobi, Mombasa, and Laikipia contexts — each requiring the identification of two or more named sub-regions (annular sector, segment, common region, full sector) within a single combined figure.
What did I learn?	Learners can decompose a composite circle-part figure into its constituent sub-regions, select and apply the correct formula for each region, combine the sub-areas using addition or subtraction, and express the solution as a structured multi-step calculation —

	skills directly needed for the Laikipia farmer's complete field-planning challenge.
How does this explain the phenomenon?	Learners articulated why each sub-region formula is selected (based on shape structure, not memorisation), peer-assessed solutions using KICD rubric language, and connected every composite shape to a specific real-life scenario in Kenya, demonstrating that the driving question — how to calculate the exact area of any part of a circle — can now be answered for complex, combined shapes.

LESSON 8 (80 minutes): Assessment, Model Revision and Consolidation — Answering the Farmer's Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners demonstrate mastery of all circle-part area formulae through a written test, then collaboratively revise their cumulative diagrammatic model to produce a final annotated explanation of how the Laikipia farmer can plan every sub-region of the irrigation field.
Knowledge	– Recall and state from memory the formulae for the area of an annulus, sector, annular sector, segment, and common region between two intersecting circles. – Explain the derivation logic connecting each formula to the base formula $A = \pi r^2$. – Identify which formula applies to each physical sub-region of the Laikipia centre-pivot irrigation field.
Skills	– Apply multi-step composite-area calculations accurately, selecting the correct formula for each sub-region. – Construct and annotate a cumulative diagrammatic model that integrates all five circle-part formulae with real-life Kenyan applications. – Communicate mathematical reasoning clearly in written test responses and group model presentations.
Attitudes	– Appreciate that mathematical precision directly prevents resource waste for Kenyan farmers, engineers, and artisans. – Show confidence in applying a connected set of formulae rather than memorising isolated rules. – Demonstrate integrity by working independently during the test and collaboratively during model revision.
Key Inquiry Question	How do we determine the area of parts of a circle, and how are these areas applied in real-life situations such as the Laikipia farmer's irrigation planning?
Purpose in Storyline	This is the culminating lesson: learners answer the driving question in full by completing a written assessment that tests every formula derived across Lessons 1–7, then produce and present a final annotated model showing the Laikipia farmer exactly how to calculate the area of every sub-region — the ring between pipes, the pivot's swept wedge, the annular-sector band, the unsown segment beside a drainage ditch, and the overlapping patch shared with a neighbouring plot.
Safety Notes	No specific hazards. Ensure learners work independently and in silence during the written test phase. Calculators and mathematical tables are permitted assessment resources per KICD guidelines.

B. LESSON OVERVIEW

Lesson 8 is the capstone of the Sub-Strand 2.6 unit. It unfolds in two clearly separated halves. In the first half (approximately 35 minutes), learners complete a class written test covering all five circle-part area concepts: annulus (Lesson 1), sector (Lesson 2), annular sector (Lesson 4), segment (Lesson 5), and common region between two intersecting circles (Lesson 6). The test is contextualised entirely within the Laikipia centre-pivot irrigation phenomenon and Kenyan real-life settings — a chapati pan border, a Kisumu roundabout sector, a Rift Valley tea-estate sprinkler overlap — so that assessment items directly mirror the anchoring phenomenon. This ensures that the assessment is not merely a recall exercise but a genuine demonstration of competency-based understanding.\n\nIn the second half (approximately 35 minutes), learners return to their cumulative diagrammatic models — the annotated aerial-view sketches of the Laikipia farm that have been revised after each lesson. Working in groups, they perform a final revision: adding any missing sub-regions, correcting formulae in light of test feedback, annotating each region with its formula and a real-life Kenyan application, and writing a one-sentence answer to the driving question on the model itself. Groups then share their final models through a brief gallery walk, and the teacher leads a whole-class consolidation that maps every formula back to the driving question and the original anchoring phenomenon. The lesson closes with learners updating the Driving Question Board one final time — moving all remaining sticky notes from 'Questions We Still Have' to 'Questions We Can Now

Answer' — and the teacher delivering a formal, concise statement of the final explanation.\n\nThroughout, the teacher uses targeted formative assessment during the gallery walk to identify any persistent misconceptions (e.g., subtracting radii before squaring in the annular sector, or omitting the triangle subtraction in the segment formula) and addresses them explicitly during consolidation. The lesson embeds NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 8 (Obtaining, Evaluating, and Communicating Information), translated here into mathematical modelling and evidence-based explanation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before the test begins, each learner individually re-examines the aerial photograph of the Laikipia centre-pivot farm (printed on the test cover sheet) and spends 3 minutes writing in their exercise books: 'Which sub-region of the farm do you think is hardest to calculate, and why?' They also predict, without calculating, whether the area of the annular sector (the swept ring) is larger or smaller than the area of the segment cut off by a straight drainage ditch. This activates prior knowledge of all five formulae and surfaces any lingering uncertainty before the test.	Distribute test papers face-down. Say: 'Before you turn over the paper, look at the aerial photograph on the cover. In your exercise book, write the name of the sub-region you think is most difficult to calculate and give one reason. You have exactly 3 minutes — go.' Start a visible timer. After 3 minutes, say: 'Now add one more sentence: predict whether the annular sector area or the segment area is larger for the same radius and angle — do not calculate, just reason.' Allow 90 seconds. Cold-call 3 learners at random to share predictions aloud (10-second wait time after each name before prompting). Say: 'Hold that prediction in your mind — your test will tell you if you were right. Turn over your papers now.'	Activation Prediction with Metacognitive Check — learners name their own area of uncertainty before assessment, which primes retrieval of all five formulae and signals to the teacher which concepts are most fragile entering the test. This is not graded; it is a low-stakes mental warm-up.	Teacher listens to the 3 cold-called predictions and notes: (1) Do learners name the correct formula for each sub-region? (2) Is reasoning about annular sector vs. segment based on formula structure (e.g., 'annular sector has $R^2 - r^2$ so it depends on both radii') or on vague intuition? This informs which consolidation points need greatest emphasis in the post-test phase.
Observe Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12	Learners complete the 35-minute class written test independently and in silence. The test contains five questions, one per formula, all set in Kenyan real-life contexts drawn from the anchoring phenomenon and local settings. Q1: The outer irrigation pipe on the Laikipia farm has radius R = 45 m and the inner pipe has radius r = 30 m. Calculate the area of the full annular ring (annulus) available for the new maize belt. Q2: After a morning's work, the centre-pivot	Circulate silently throughout the test, scanning for procedural errors without intervening. Specifically watch for: (a) learners who subtract radii before squaring in Q3 — note their names; (b) learners who omit the $½r^2\sin θ$ triangle subtraction in Q4 — note their names; (c) learners who use the wrong angle (not 2θ) in Q5 — note their names. At 25 minutes, quietly announce: 'Ten minutes remaining — check that you have shown all working and included	Contextualised Summative Evidence Gathering — all five test questions are embedded in the Laikipia phenomenon and Kenyan settings. This means learners are not just recalling formulae in abstract; they are demonstrating that they can decompose the real farm scenario into the correct sub-region type and execute the calculation. The test serves as structured 'observed evidence' that each formula correctly describes a measurable physical region.	After collecting papers, teacher immediately reveals the model answers on the board (or reads them aloud): Q1: $π(45^2−30^2) = π(2025−900) = 1125π ≈ 3534.29$ m²; Q2: $(135/360)×π×45^2 = (3/8)×2025π = 759.375π ≈ 2386.00$ m²; Q3: $(135/360)×π×(45^2−30^2) = (3/8)×1125π ≈ 1325.36$ m²; Q4: $(80/360)×π×20^2 − ½×20^2×\sin 80° = (2/9)×400π − 200×0.9848 ≈ 279.25 − 196.96 ≈ 82.29$ m²; Q5: $\cos α = (25^2+30^2−25^2)/(2×25×30) =$

🔍 Search ARES for similar readings	arm has swept through 135°. Using r = 45 m, calculate the area of the sector watered. Q3: The farmer adds a second arm at r = 30 m. Calculate the area of the annular sector swept through 135° between r = 30 m and r = 45 m. Q4: A straight drainage ditch cuts a chord across the field at a perpendicular distance creating a central angle of 80° in a circle of radius 20 m. Calculate the area of the segment between the ditch and the arc. Q5: Two adjacent circular plots in the Rift Valley both have radius 25 m and their centres are 30 m apart. Calculate the area of the overlapping common region. Learners are permitted calculators and mathematical tables.	units in every answer.' At 33 minutes: 'Two minutes — finalize your responses.' At 35 minutes: 'Pens down. Pass papers to the front of your row.' Collect all papers.		900/1500 = 0.6, α = 53.13°; each segment = (106.26/360)×π×25² − ½×25²×sin106.26° ≈ 579.56 − 300.05 ≈ 279.51 m²; total ≈ 559.02 m². Learners self-mark using the model answers. Teacher notes class-wide error patterns for consolidation.
Explain Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After self-marking, each learner writes a 'Formula Passport' — a half-page table in their exercise book listing all five formulae, a one-sentence derivation rationale, and one Kenyan real-life application for each. They then return to their groups (same groups as Lessons 1–7) and complete a final revision of their cumulative diagrammatic model of the Laikipia farm. Each group annotates the aerial-view sketch with: (1) the name of each sub-region clearly labelled, (2) its formula written beside it, (3) the numerical answer from their test cross-checked against the group consensus, and (4) a one-sentence real-life interpretation for the farmer (e.g., 'The annular sector of 1325.36 m² tells the farmer how much fertiliser to budget for the outer ring when the arm has swept 135°.'). Groups also write a one-sentence answer to the driving question directly on	After model answers are reviewed, say: 'You have 5 minutes individually — complete your Formula Passport in your exercise book now.' Circulate and confirm every learner includes all five formulae. After 5 minutes: 'Now move into your groups. Your task is to complete the final revision of your cumulative model. Every sub-region must be labelled, every formula written in, every number checked, and the driving question answered on the poster. You have 12 minutes.' During group work, stop at each group and ask one targeted question: To Group 1 — 'Why do we square R and r separately before subtracting in the annular sector formula?' (Expect: because area scales as r², not r; subtracting radii first would give the wrong geometry.) To Group 2 — 'What two areas do we subtract to get the segment area, and why?' To Group 3 — 'How did you use the cosine rule to	Formula Passport + Annotated Model Integration — the Formula Passport forces each learner to individually consolidate all five formulae in one organised record before group work begins, preventing social loafing. The annotated model then requires groups to connect each formula to a physical region and a practical Kenyan application, operationalising NGSS Practice 6 (Constructing Explanations) in a mathematical modelling context.	Teacher checks Formula Passports for completeness and correct derivation language (e.g., 'Segment = Sector − Triangle' not just 'subtract something'). During group annotation, teacher listens for correct use of terms: annulus, sector, annular sector, segment, common region. Red flag: any group that writes the annular sector formula as (θ/360)×π×(R−r)² — teacher intervenes immediately with the question: 'If I double both R and r, does the ring area double or quadruple?' to expose the error.

	the model poster.	find the angle for the common region?' Allow 10-second wait time per question before offering prompts. At 12 minutes: 'Finish your driving-question sentence now and pin your model on the wall for the gallery walk.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Sector (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners conduct a 6-minute silent gallery walk, moving from poster to poster and placing a green dot sticker on any formula annotation they find particularly clear, and an orange dot on any they find confusing or incomplete. Each learner also reviews the Driving Question Board at the front of the room — the large board that has accumulated sticky-note questions since Lesson 1. They pick up a final sticky note and write either: (a) 'ANSWERED: [question] — because [one-sentence reason]' and move it to the 'Questions We Can Now Answer' column, or (b) 'STILL WONDERING: [new question that goes beyond this unit]' and post it in a new column labelled 'Extensions'. The class then assembles at the DQB for a 5-minute whole-class share.	Say: 'You have 6 minutes for the gallery walk. Green dot = this annotation is clear and correct. Orange dot = I am unsure about this or it seems incomplete. Move in silence and keep moving.' Use a timer. After the gallery walk, direct learners to the DQB: 'Look at the questions we posted since Lesson 1. Take a sticky note. If you can now answer a question that was unresolved, write ANSWERED and move it. If this unit has raised a new question for you — maybe about 3D shapes, or computer-aided design for irrigation — write STILL WONDERING and put it in the Extensions column.' Allow 3 minutes. Then cold-call 4 learners (one per table group) to read their sticky note aloud and explain their reasoning. 10-second wait time per learner. Say: 'Notice that our DQB has shifted — most of the original questions are now in the ANSWERED column. That is what learning a unit of mathematics looks like.'	Gallery Walk with Dot Voting + DQB Final Audit — the gallery walk gives learners peer feedback on their own model quality without the teacher as sole arbiter, building metacognitive awareness of what strong mathematical annotation looks like. The DQB final audit makes visible the cumulative progression of understanding across all 8 lessons, helping learners see their own intellectual growth rather than experiencing the unit as a series of disconnected lessons.	Teacher counts the proportion of DQB sticky notes successfully moved to 'ANSWERED'. Target: ≥ 90% of original questions answered. Teacher reads all 'STILL WONDERING' notes after class to identify genuine extensions (e.g., surface area of a sphere sector, use of integration for irregular areas) that could seed future units or enrichment tasks. Orange dots on gallery-walk models identify which group's annotation needs direct teacher follow-up during the consolidation phase.
Model Building Phase 📹 **VIDEO:** Intercepts from an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The teacher leads a 10-minute whole-class final consolidation using one selected group's model as the anchor (the model judged by gallery-walk green dots to be most complete). The class compares this final model with a projected image of a Lesson 1 model (showing only the annulus and the basic circle). Learners articulate, in full sentences, what	Project the Lesson 1 model beside the best final model. Say: 'What did we know after Lesson 1? What do we know now? I want three volunteers to name one thing that was added.' Cold-call 3 learners with 10-second wait time. Accept responses and build toward the final explanation. Then say: 'I am going to read our final explanation aloud. Follow in your books and	Lesson 1 vs. Final Model Comparison (Cumulative Model Revision) — placing the Lesson 1 model beside the final model makes the entire learning trajectory visible in a single image, a key sensemaking move for helping learners consolidate a multi-lesson unit into a coherent schema. The formal Final Explanation statement	Teacher scans final personal diagrams for: (1) all five sub-regions present and correctly named; (2) formulae written correctly — especially $A = \pi(R^2 - r^2)$ for annulus, $A = (\theta/360)\pi r^2$ for sector, $A = (\theta/360)\pi(R^2 - r^2)$ for annular sector, $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ for segment, and the two-segment sum for the common region; (3) at least one numerical

📖 **HTML:** [Area of a Sector (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	has been added across Lessons 1–8. The teacher then writes the Final Explanation on the board as a formal class statement: 'The area of any part of a circle can be calculated by (1) identifying whether the region is an annulus, sector, annular sector, segment, or common region; (2) applying the correct formula; and (3) summing or subtracting sub-regions for composite shapes. Using these formulae, the Laikipia farmer can calculate the exact area of every irrigated sub-region — the ring between pipes, the swept wedge, the annular band, the unsown segment, and the shared overlap with a neighbour's plot — enabling precise planning of water, fertiliser, seed, and fencing budgets with no waste.' Each learner copies this final explanation into their exercise book and draws a final personal annotated diagram of the Laikipia farm, labelling all five sub-regions with their formulae.	correct anything you had wrong.' Read the final explanation slowly and clearly. Then: 'Copy this exactly into your exercise book — it is the answer to the question we have been working on since Lesson 1.' Allow 3 minutes for copying. Then: 'Now draw your own annotated diagram of the Laikipia farm — include all five sub-regions, each with its formula and one number from today's test. You have 5 minutes.' Circulate and check that every learner includes all five regions. Close the lesson by returning to the original aerial photograph: 'This is the farmer's farm in Laikipia. Eight lessons ago we could only calculate the full circle. Today, we can calculate every part of it. That is the power of decomposing a complex shape into precise sub-regions — and it is exactly the mathematics that engineers, architects, and farmers use in Kenya every day.'	operationalises NGSS Practice 6 (Constructing Explanations) and NGSS Practice 8 (Communicating Information), giving learners a precise, evidence-based answer to the driving question in their own records.	value from the test attached to each region. Any learner whose diagram is missing two or more regions receives a targeted follow-up task before the next unit begins. Teacher records overall test performance data against the KICD rubric descriptors (Exceeds / Meets / Approaches / Below Expectations) for 'Ability to determine the area of an annulus, sector, annular sector, and segment of a circle' and 'Ability to determine the area of a common region between two intersecting circles'.

D. TEACHER REFLECTION

1. Looking at the class written test results, which of the five formulae had the highest error rate — and was the error conceptual (wrong formula chosen) or procedural (correct formula but arithmetic error)? What does this tell me about which lessons in the sequence need to be strengthened before I teach this unit again?\n2. During the gallery walk, which group's annotated model received the most green dots, and what specific features made it clearer than the others? How can I use that model as a teaching exemplar for future cohorts?\n3. Did every learner successfully move at least one sticky note to 'ANSWERED' on the Driving Question Board? For any learner who could not, what support mechanism (peer tutoring, remedial task, digital resource on ARES) should I put in place before moving to Sub-Strand 2.7 Surface Area and Volume of Solids?\n4. When I read the 'STILL WONDERING' sticky notes, what genuine mathematical curiosity did learners express — for example, questions about 3D applications (volume of a sector-shaped silo), or digital tools (using GeoGebra to model the pivot sweep)? How might I seed these into future lessons to sustain inquiry momentum?\n5. Did the two-part lesson structure (test then model revision) give learners enough time to produce a high-quality final model, or should I allocate a full double lesson (80 minutes) for this capstone in future? Which part — the test, the model revision, or the gallery walk — felt most rushed, and how would I rebalance the timing?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed their own completed cumulative diagrammatic models of the Laikipia farm displayed side by side — showing

	the progression from the single annulus of Lesson 1 to a fully decomposed farm plan annotated with all five circle-part formulae. During the gallery walk, they also observed which annotations were rated clearest by peers (green dots) and which were flagged as unclear (orange dots), providing immediate visual evidence of which formula representations communicate most effectively.
What did I learn?	Learners consolidated all five formulae for the area of parts of a circle — annulus $A = \pi(R^2 - r^2)$, sector $A = (\theta/360)\pi r^2$, annular sector $A = (\theta/360)\pi(R^2 - r^2)$, segment $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$, and the common region as the sum of two segment areas derived via the cosine rule — and demonstrated the ability to select and apply the correct formula for each physical sub-region of a composite figure, expressing results in the correct units with structured working.
How does this explain the phenomenon?	The Laikipia farmer's planning problem — how to calculate the exact area of the ring between two irrigation pipes, the swept wedge at any point in the pivot's rotation, the annular band swept by the outer ring, the unsown region beside a drainage ditch, and the shared overlap with a neighbouring plot — is fully solved by decomposing the field into its constituent circle-part sub-regions and applying the appropriate derived formula to each. The driving question is answered: any part of a circle can be measured exactly, enabling Kenyan farmers, engineers, and artisans to plan resources precisely and avoid waste.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.